

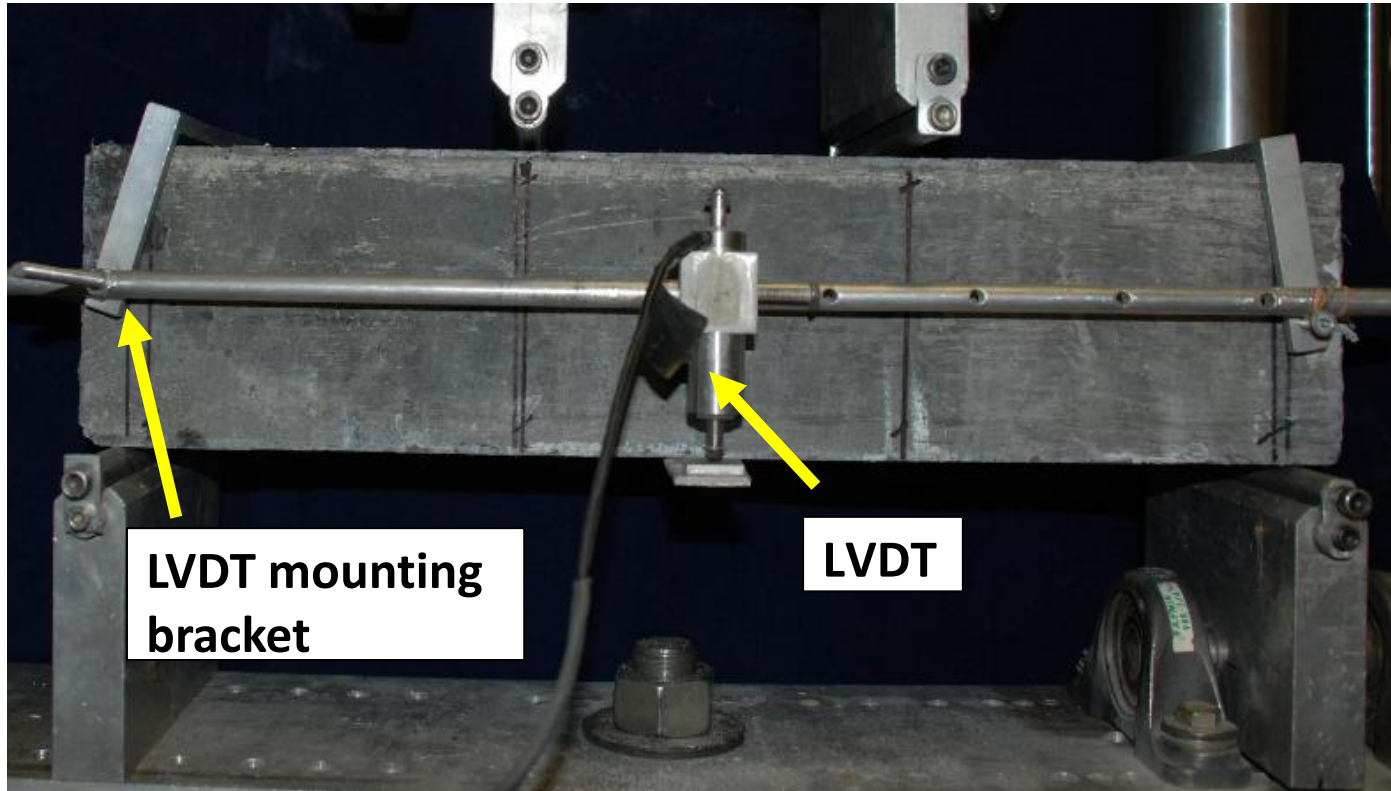
Closed-loop testing Effects on the Inverse Analysis of Tensile and Flexural Response of FRC and UHPC

Barzin Mobasher, Devansh Patel, and Chidchanok Pleesudjai

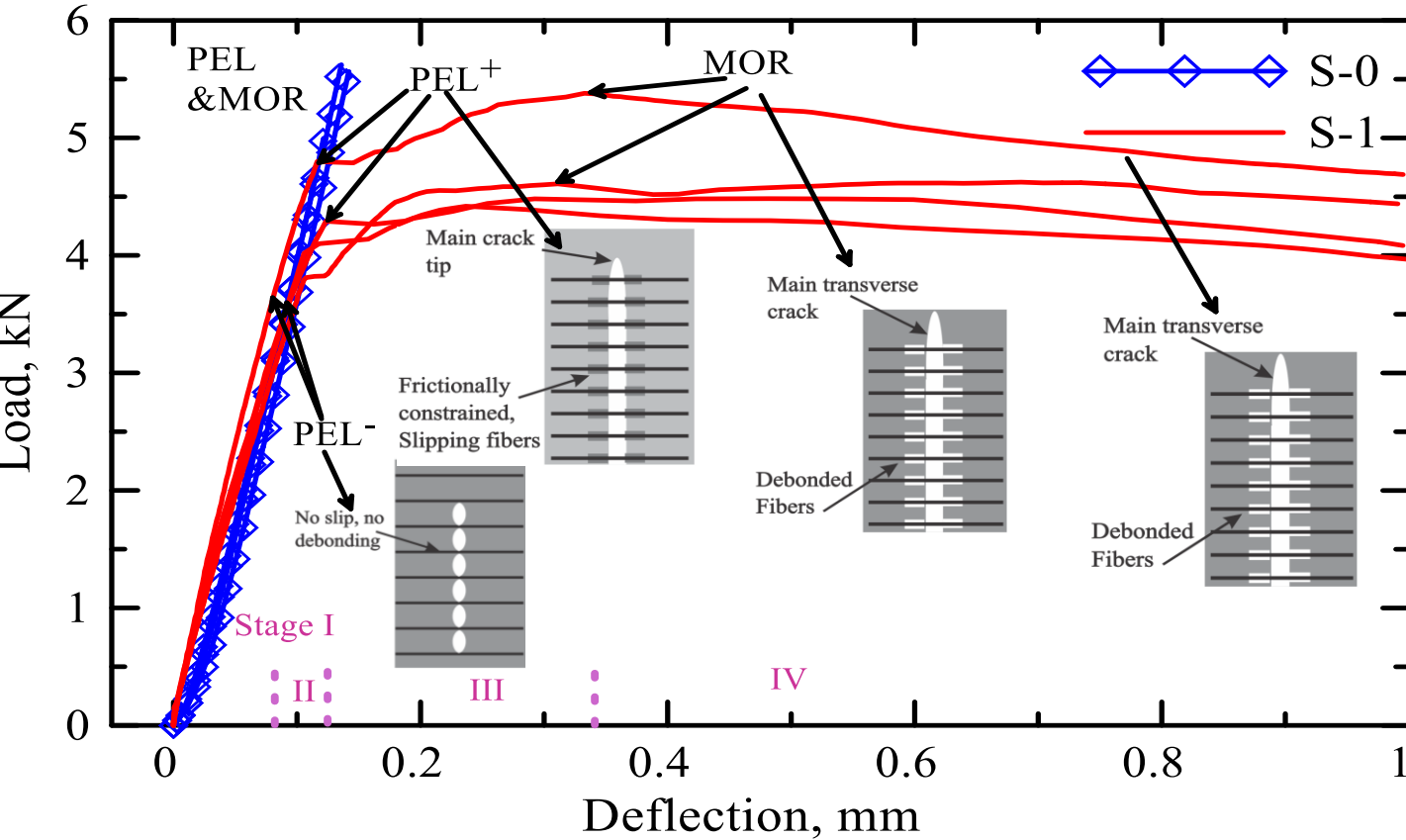
Forum on Direct Tension Testing Versus Inverse Analysis Procedures and
Bending Tests – ACI 239, ACI 544

ACI spring convention 2026

ASTM C1609, flexural beam Test



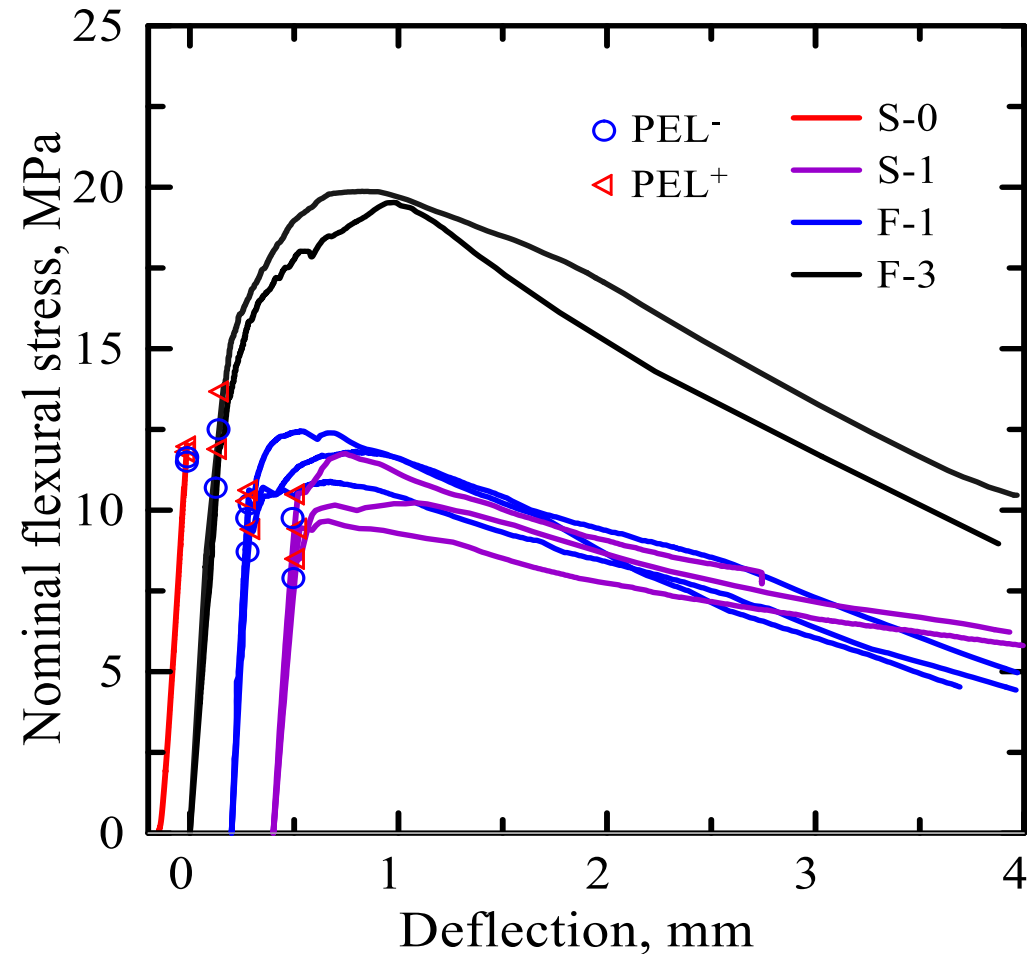
Brittleness of plain UHPC and the effect of fibers in flexure



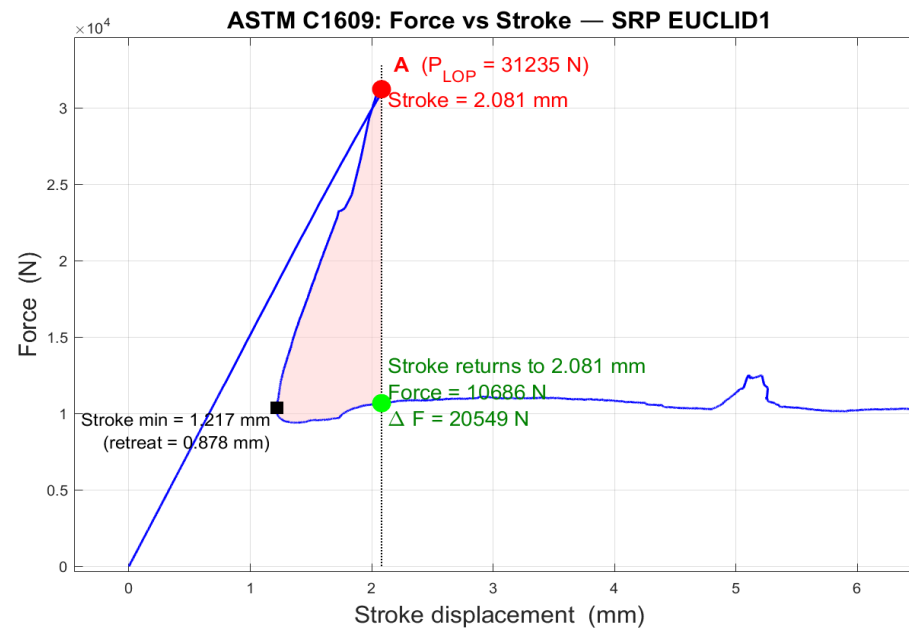
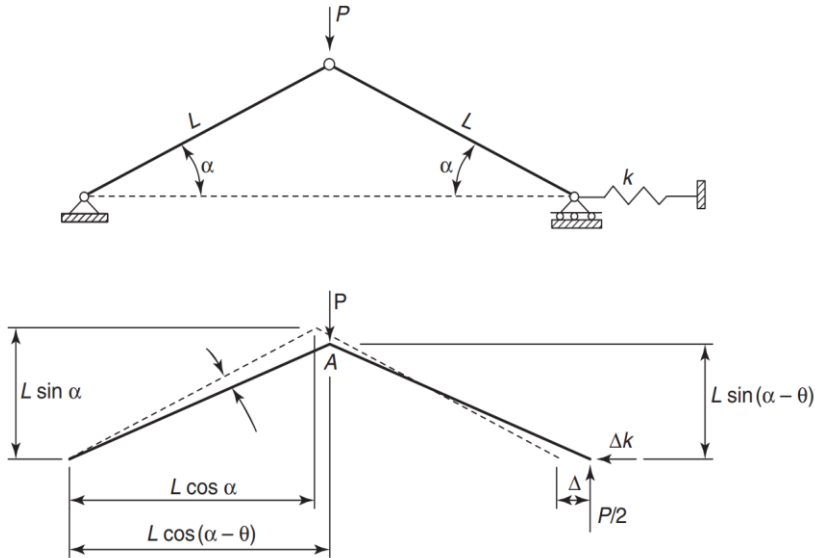
- Plain UHPC is extremely brittle
- First crack strength actually decreases when fibers are added
- Nonlinearity in the post peak and significant ductility with fibers content

Effect of fibers on the pre-peak toughening

- Initiation of cracks
- propagation of cracks,
- Localization, multiple cracking
- fiber bridging mechanisms

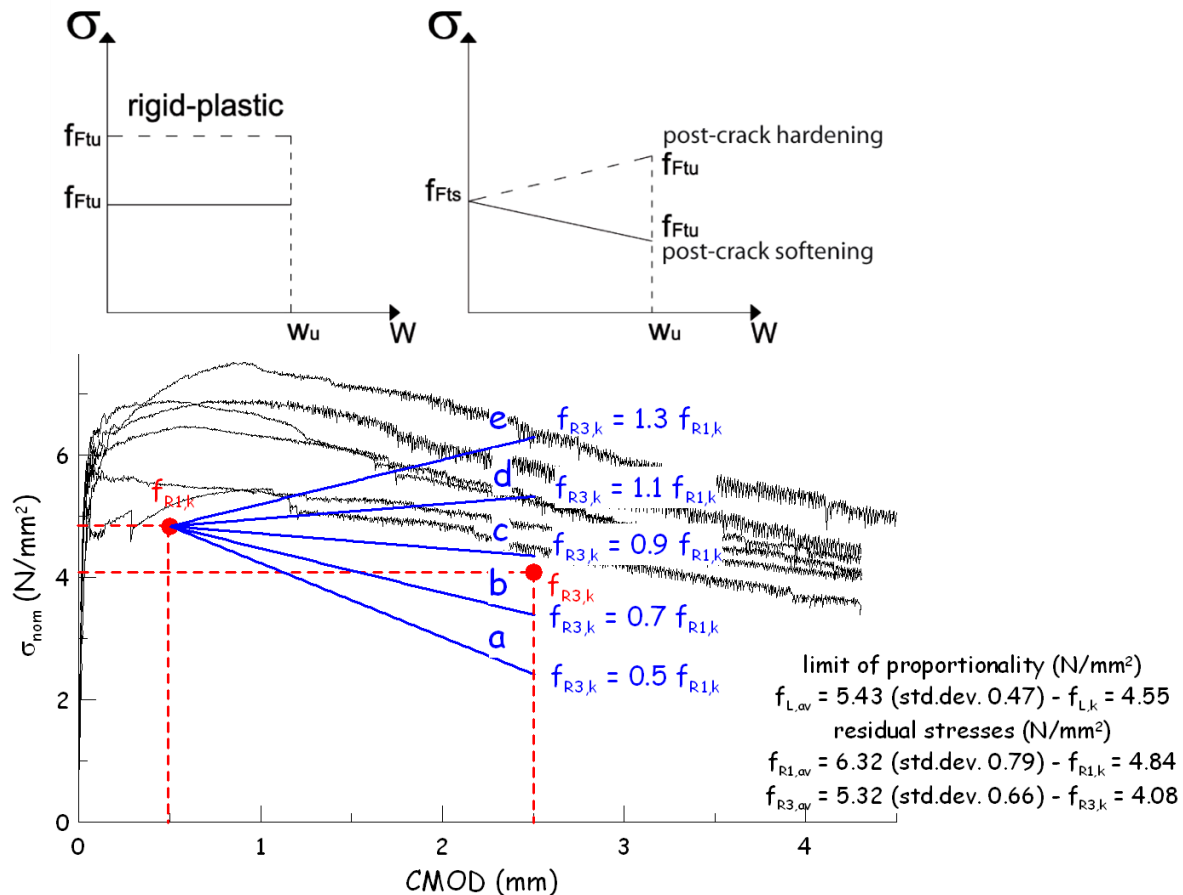


Snap-Through Instability at First Crack

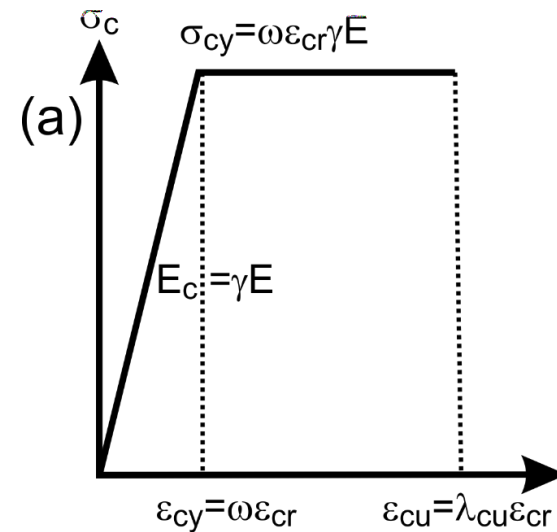


Code Approach, ASTM uses a residual Tensile Strength approach from deflection of the beam, f150

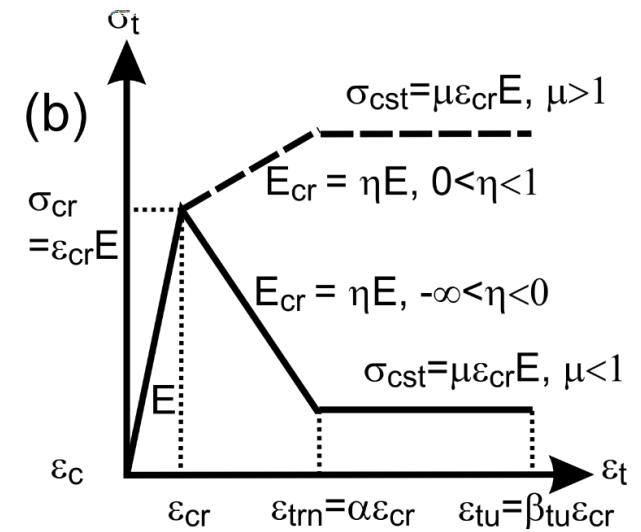
- The procedure developed by EN 14651 requires running a 3-point bending test on a notched beam, collecting the load vs. crack opening response.



Model Code Approach



Compression model



Tension model

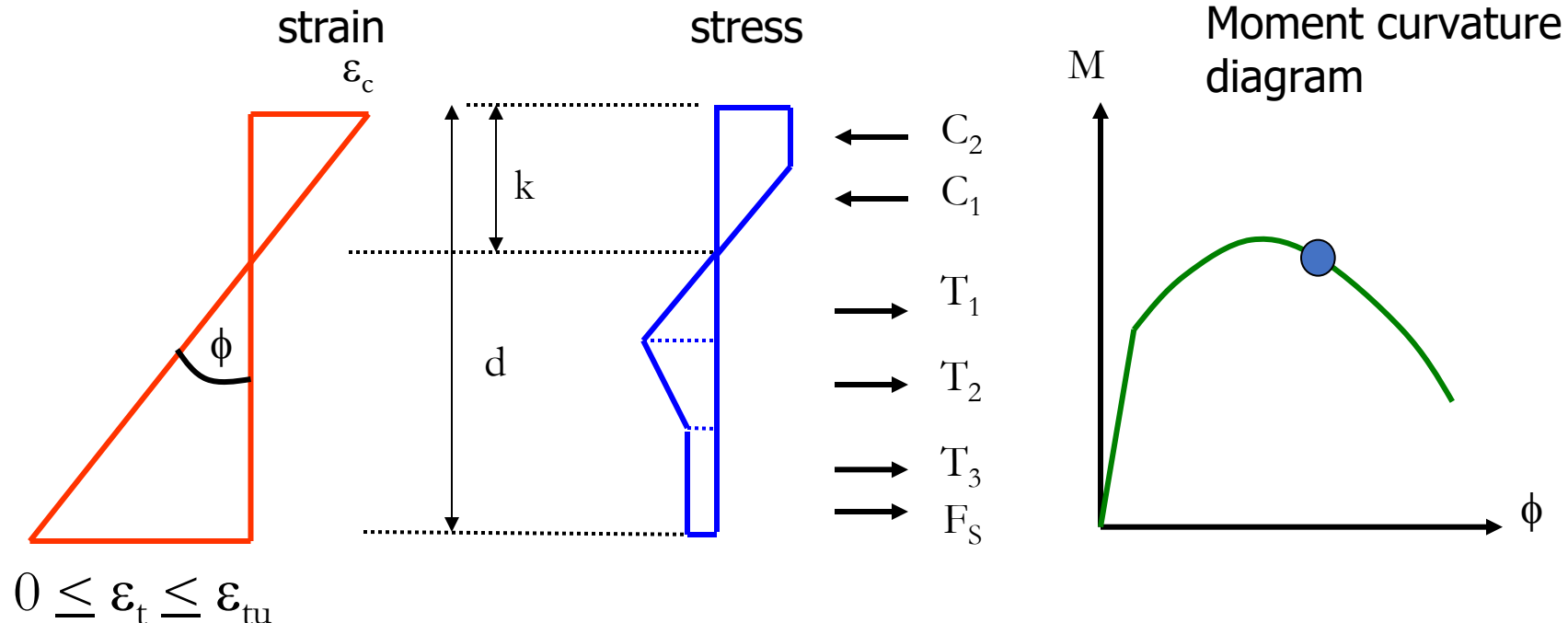
Material parameters are described as a multiple of the first cracking tensile strain (ϵ_{cr}) and tensile modulus (E)

Moment-Curvature Diagram

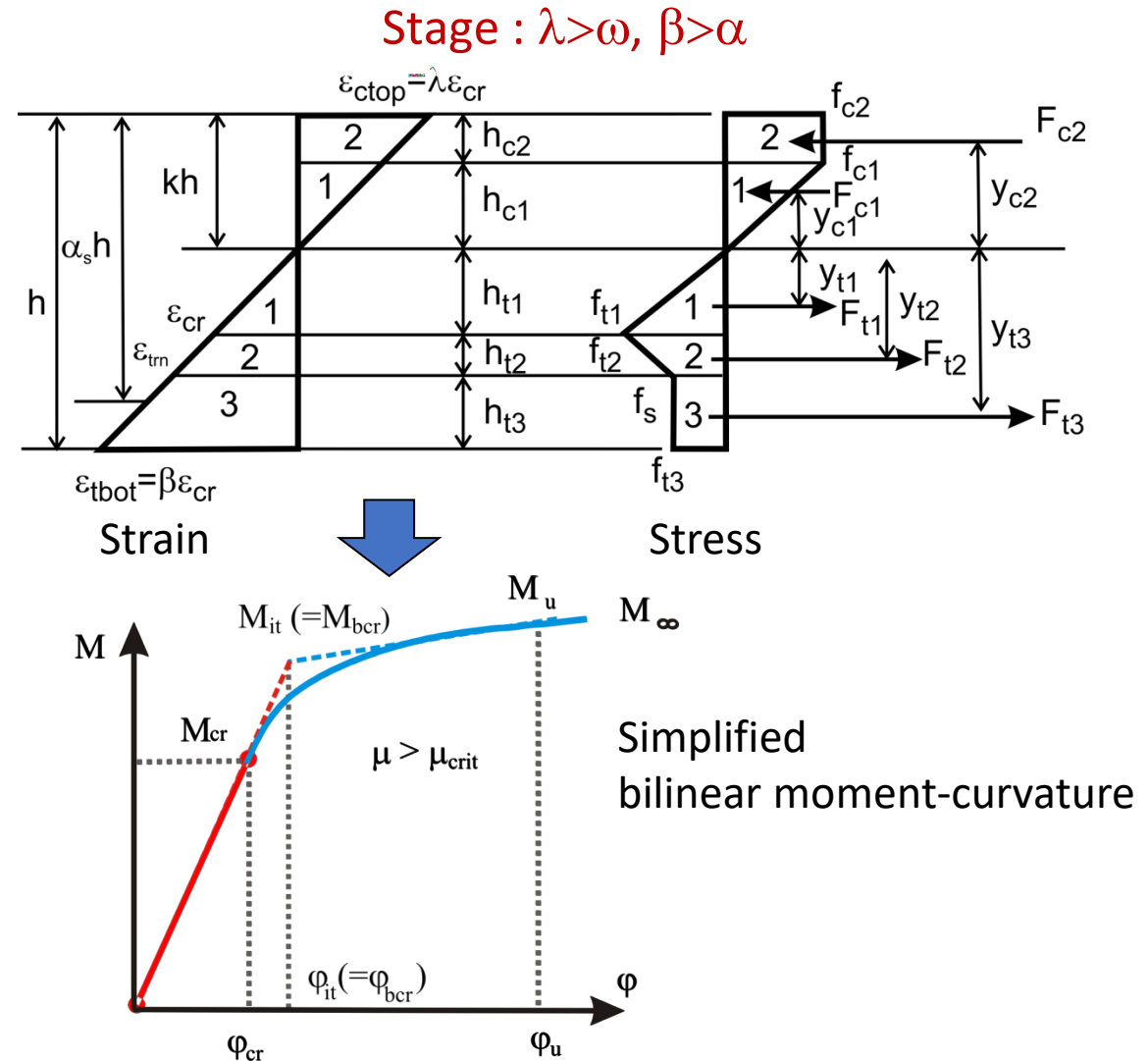
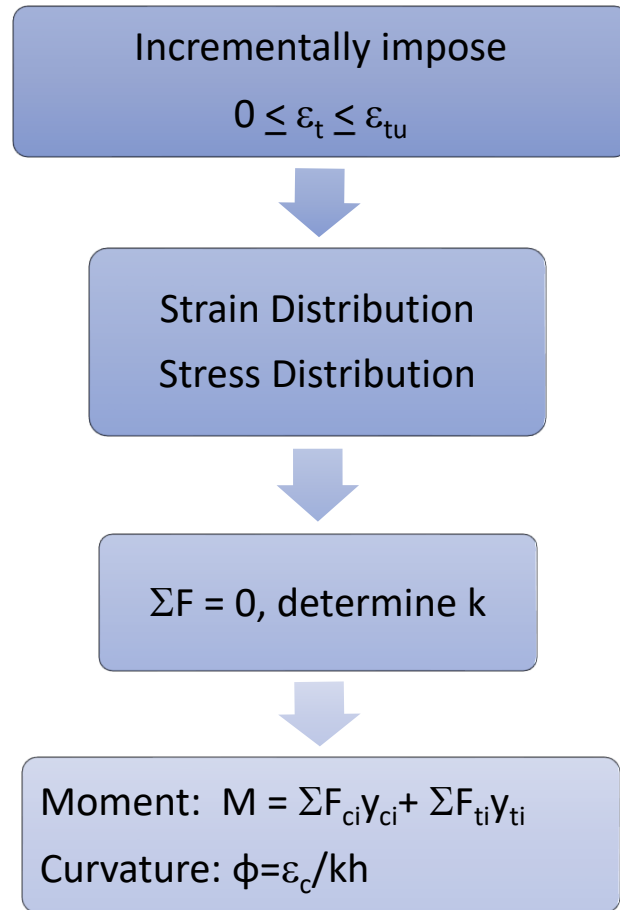
- Incrementally impose $0 \leq \varepsilon_t \leq \varepsilon_{tu}$
- Strain Distribution
- Stress Distribution
- $\Sigma F = 0$, determine k (Neutral axis)
- $M = \Sigma C_i y_{ci} + \Sigma T_i y_{ti}$ and $\phi = \varepsilon_c / kd$
- Normalization $M' = M / M_0$ and $\phi' = \phi / \phi_{cr}$

$$F_{c1} = b \int_0^{kd} f_{c1}(y) dy$$

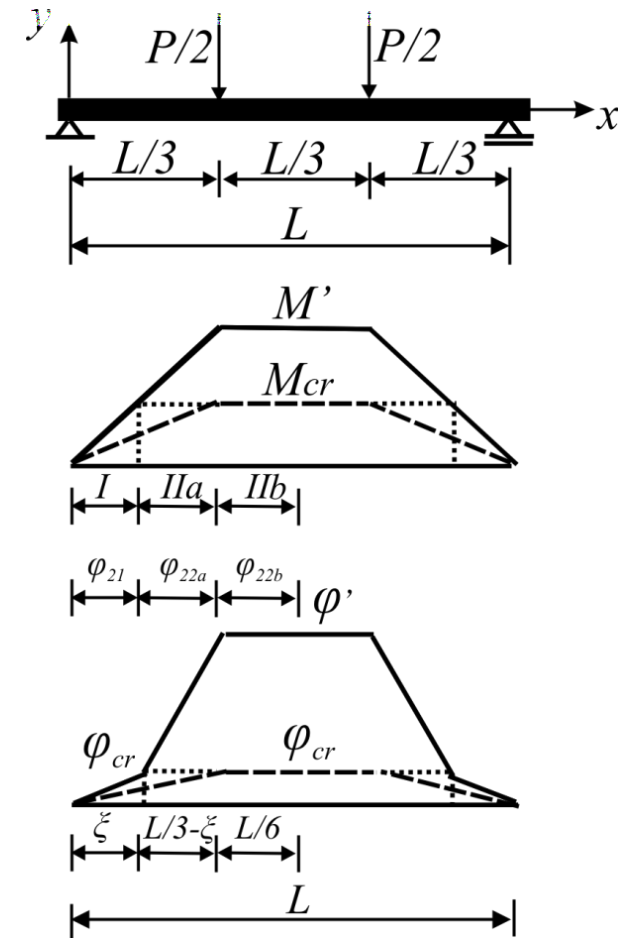
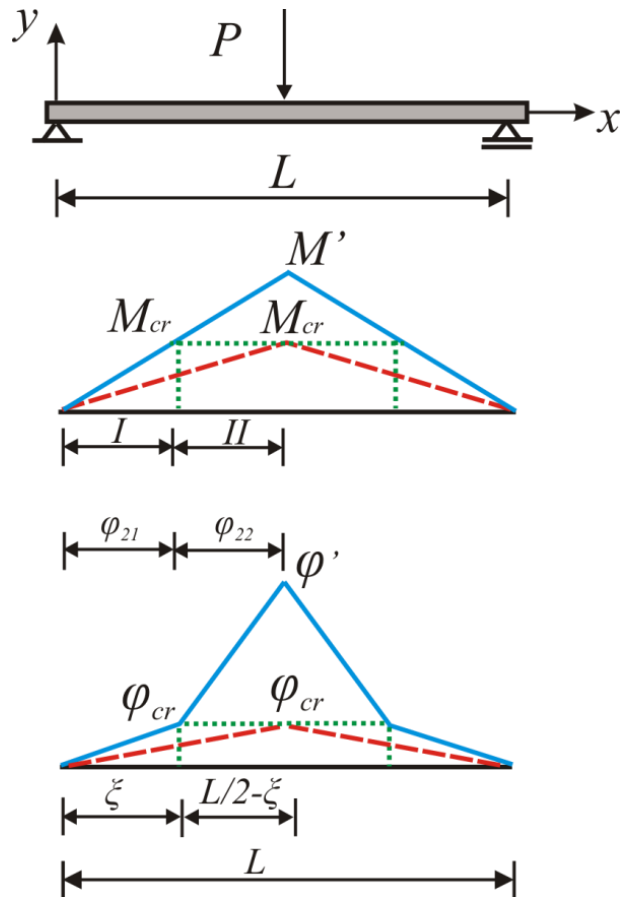
$$y_{c1} = \frac{b}{F_{c1}} \int_0^{kd} f_{c1}(y) y dy$$



Derivation of Moment-Curvature Relationship

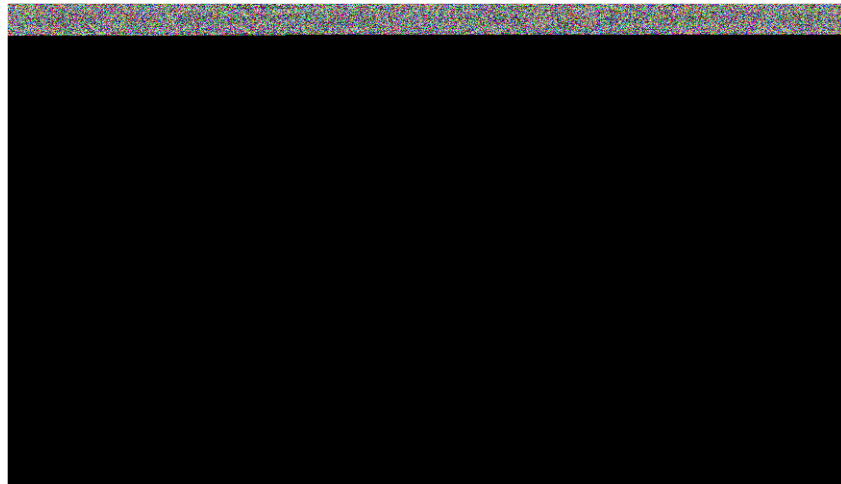
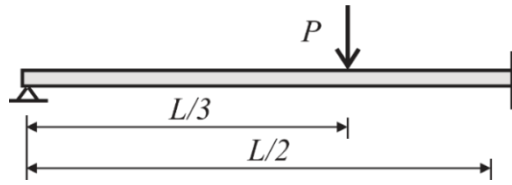


Moment-Curvature Distributions to load-Deflection

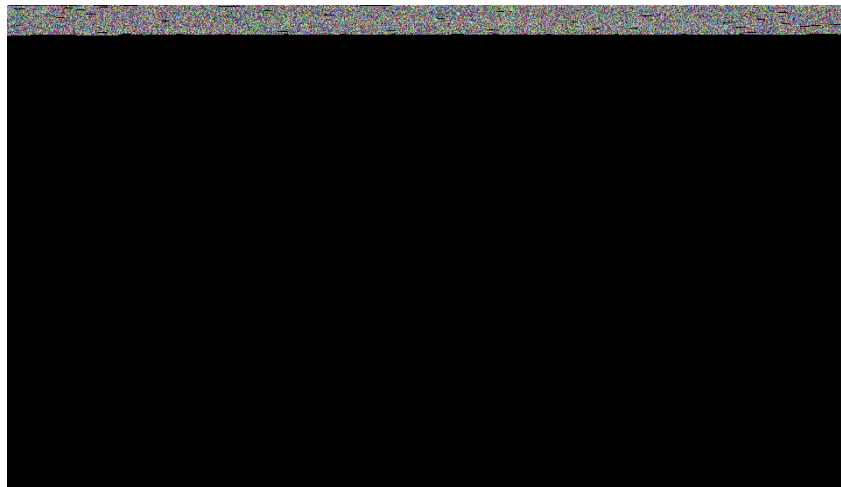


- Use Static Equilibrium to get moment distribution
- Curvature distributions along the beam are generated based on simplified bilinear moment-curvature model

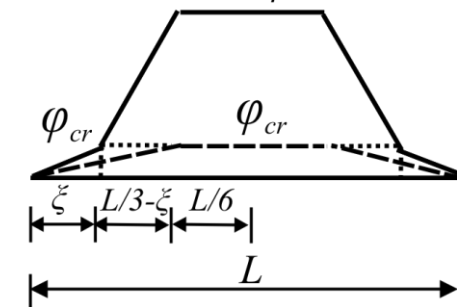
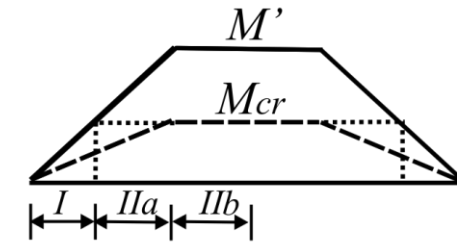
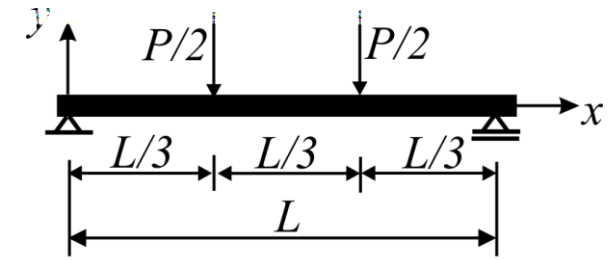
2D Contour – Distribution loading



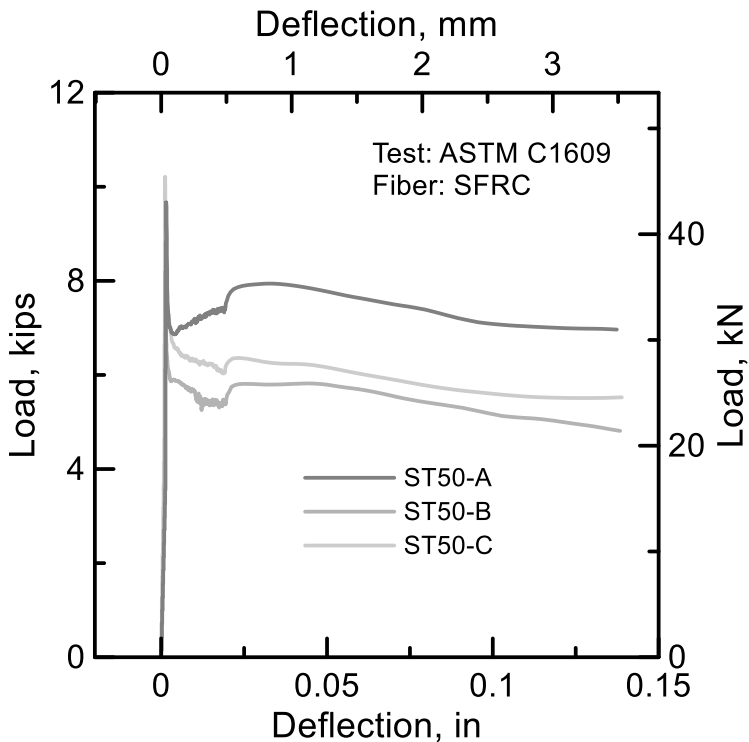
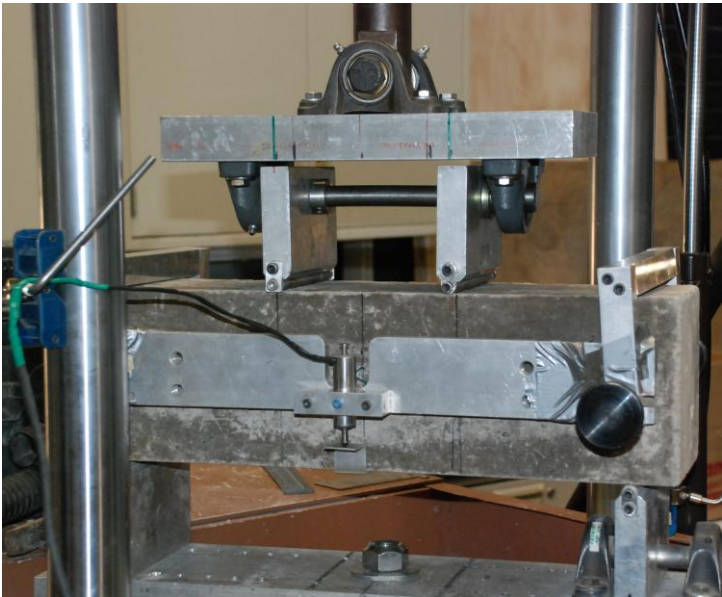
Normal strain distribution



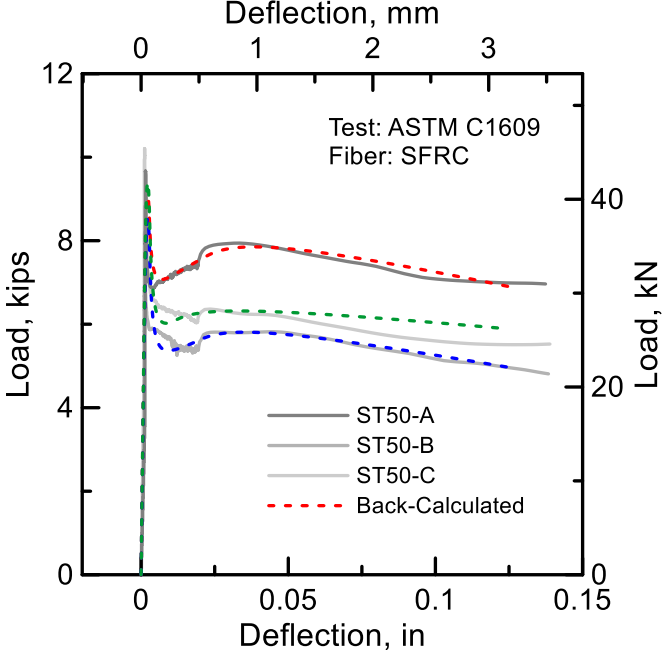
Normal stress distribution



Quadrilinear Modeling for Simulating the response of tunnel segments based on ASTM C1609 Tests

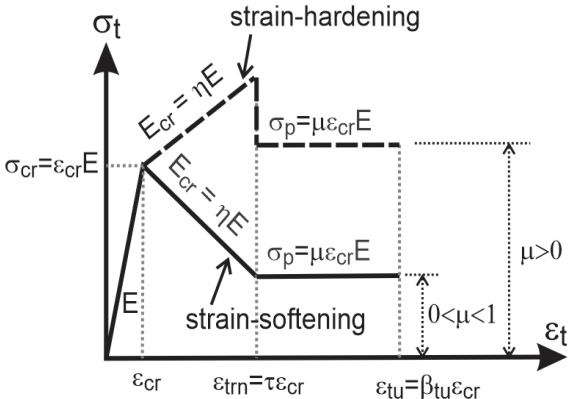
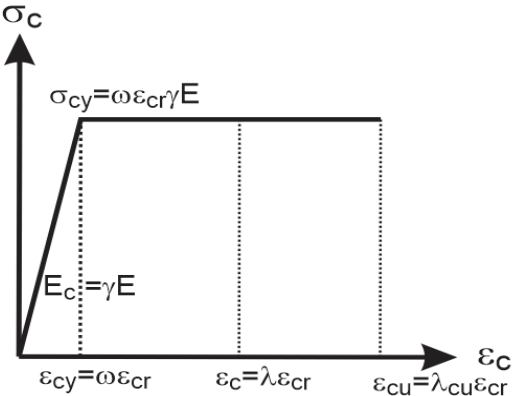


Model
Simulated
Data



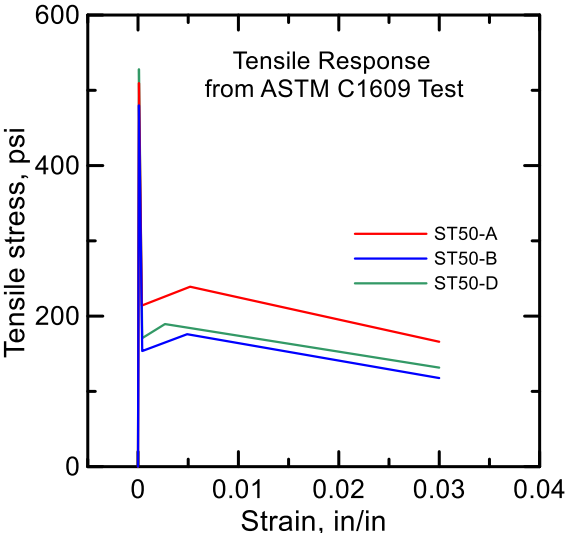
(a) Compression Model

(b) Tension Model

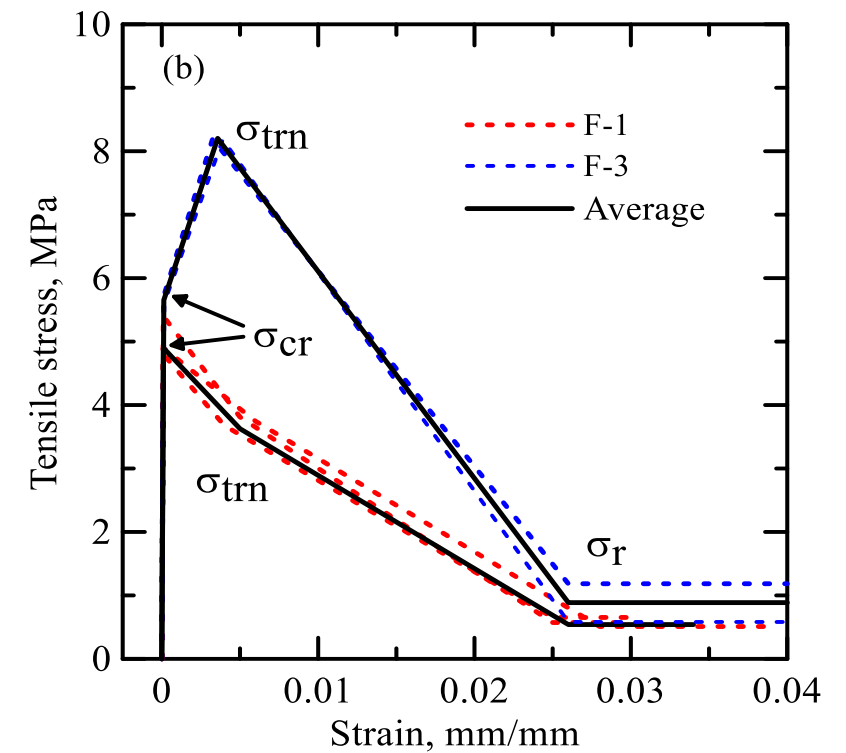
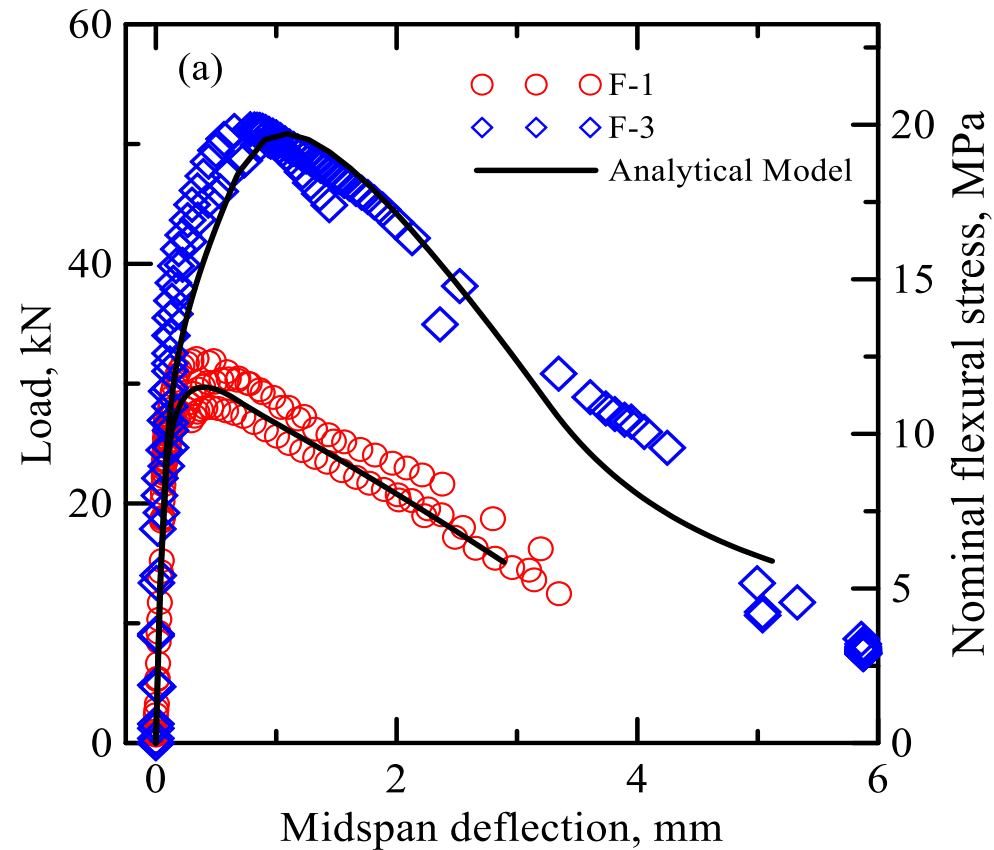


Specimen Level testing

Constitutive
Response

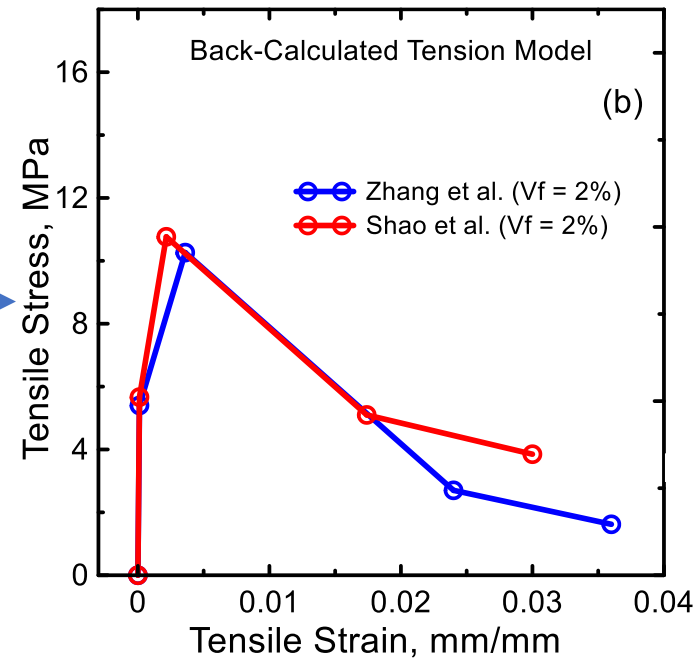
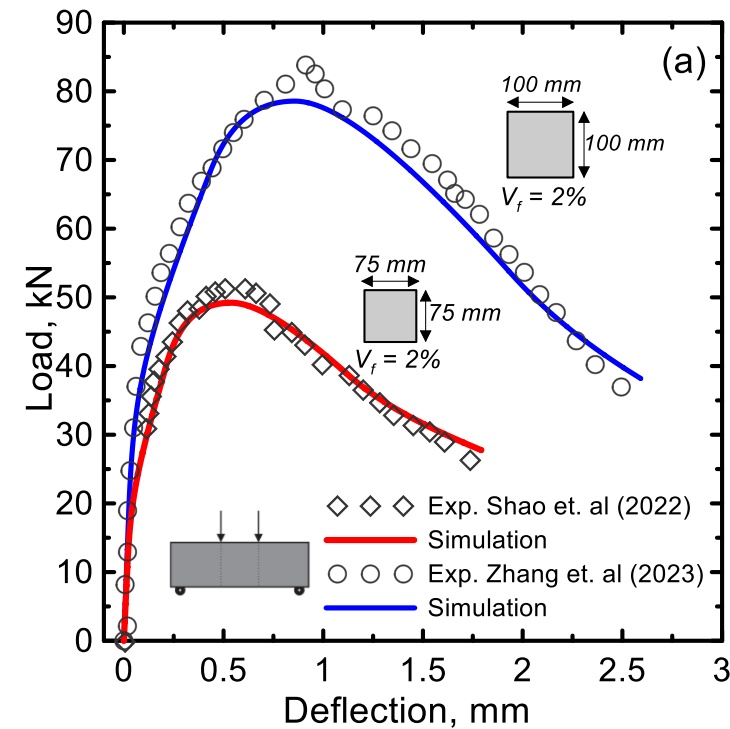


Trilinear or Quadrilinear tensile response is applicable to both FRC and UHPC

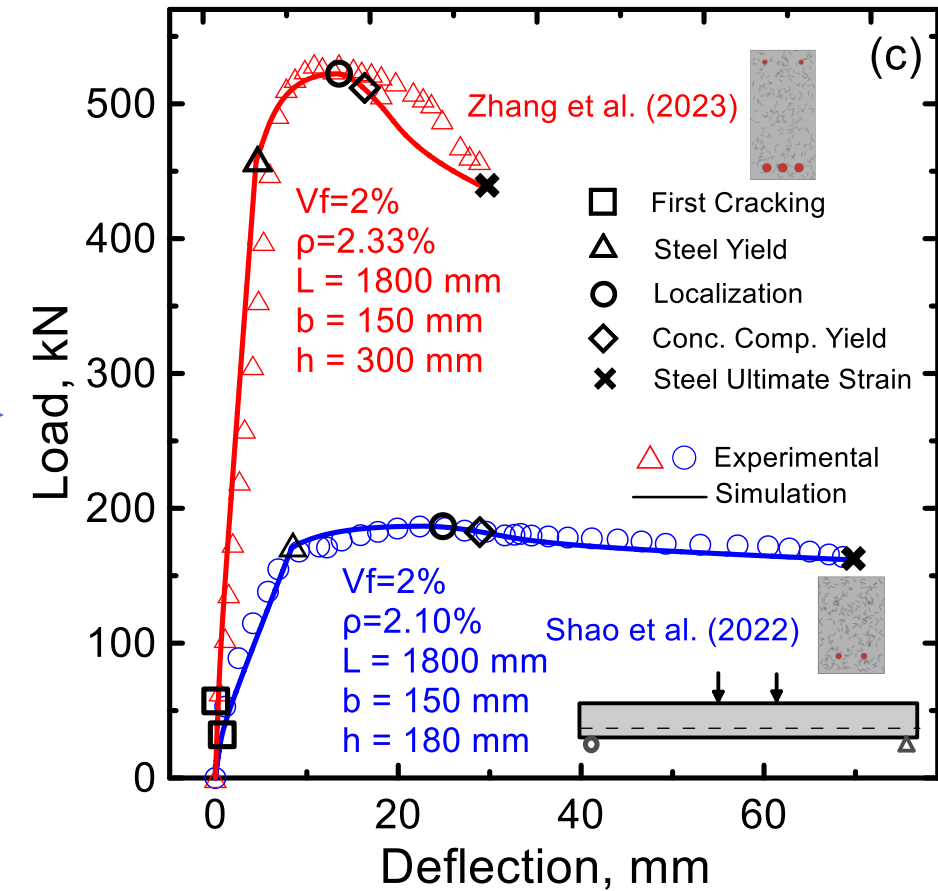


Inverse-Analysis as a reliable Characterization Approach

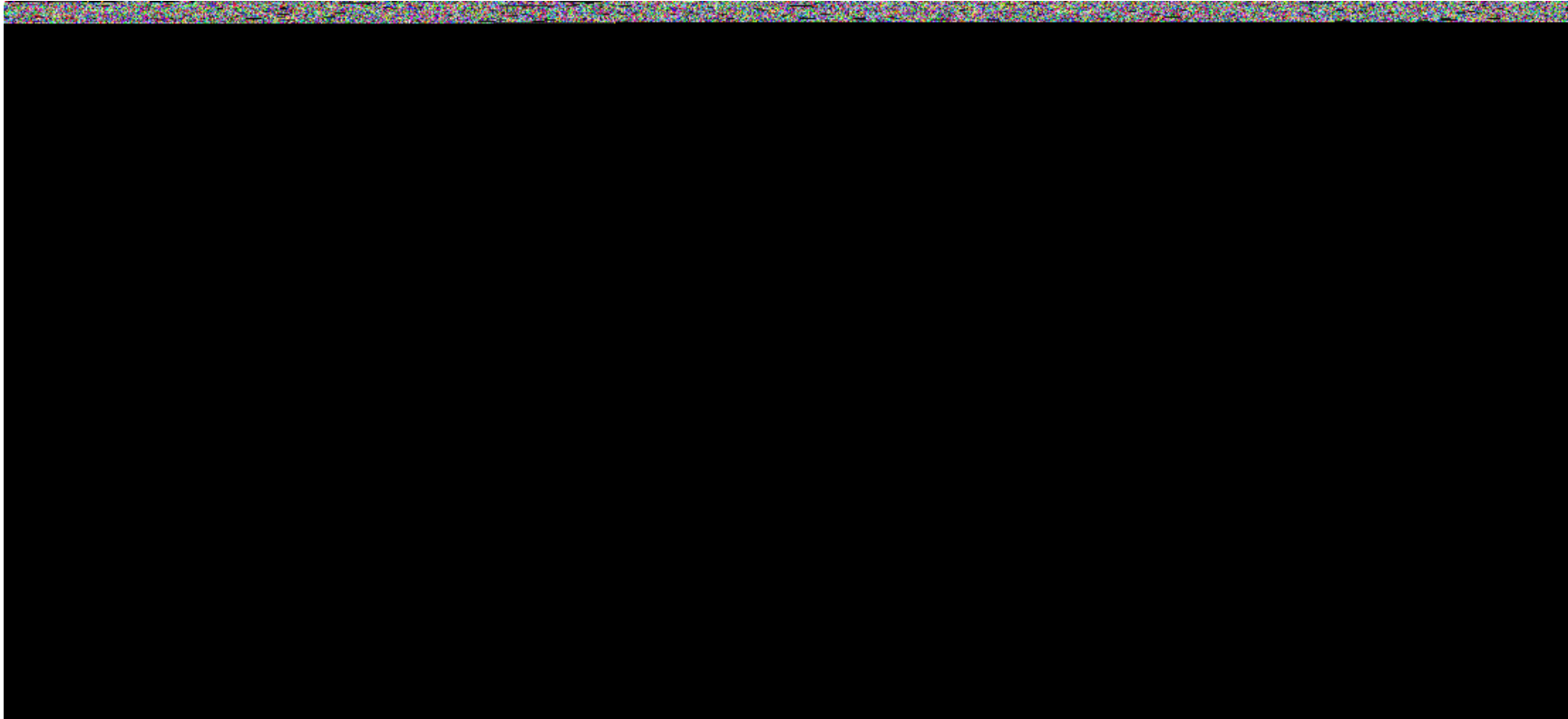
Inverse-analysis from two separate publication of different size of standard beam tests, same V_f



Scaling to R-UHPC Members With varying depths



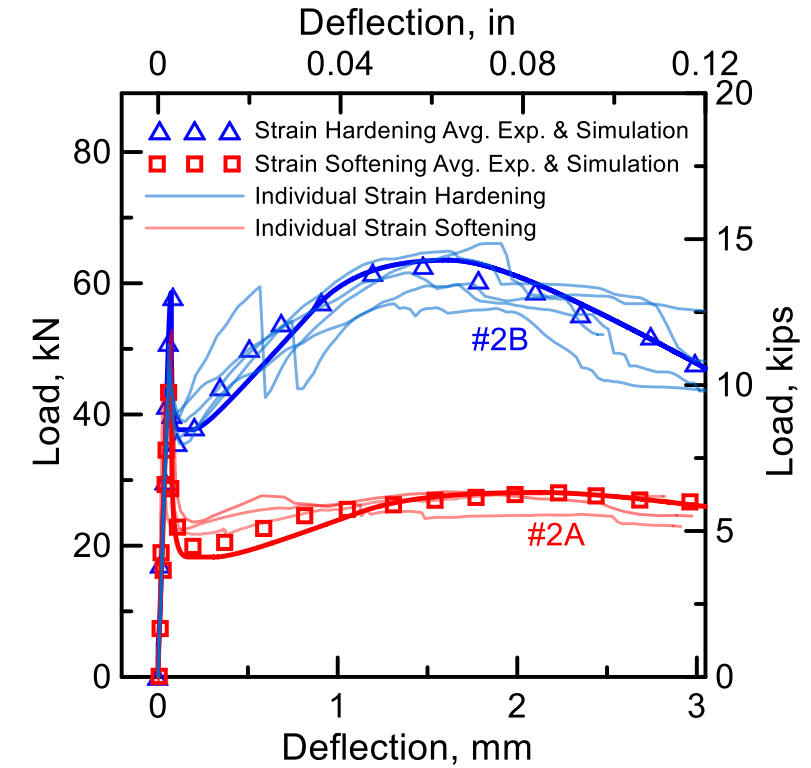
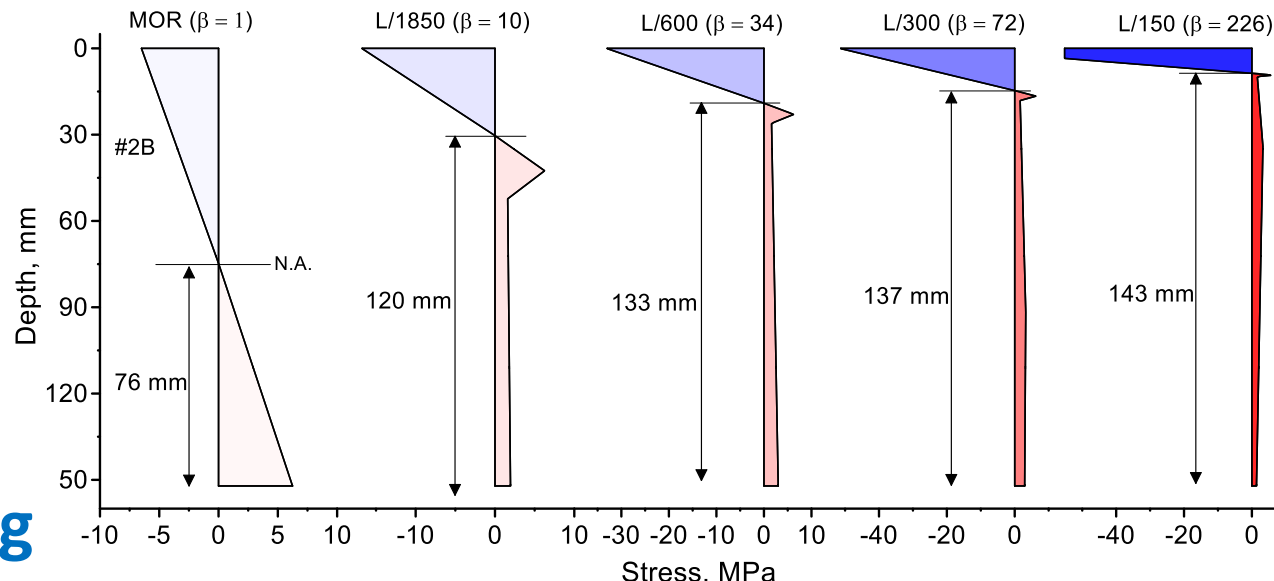
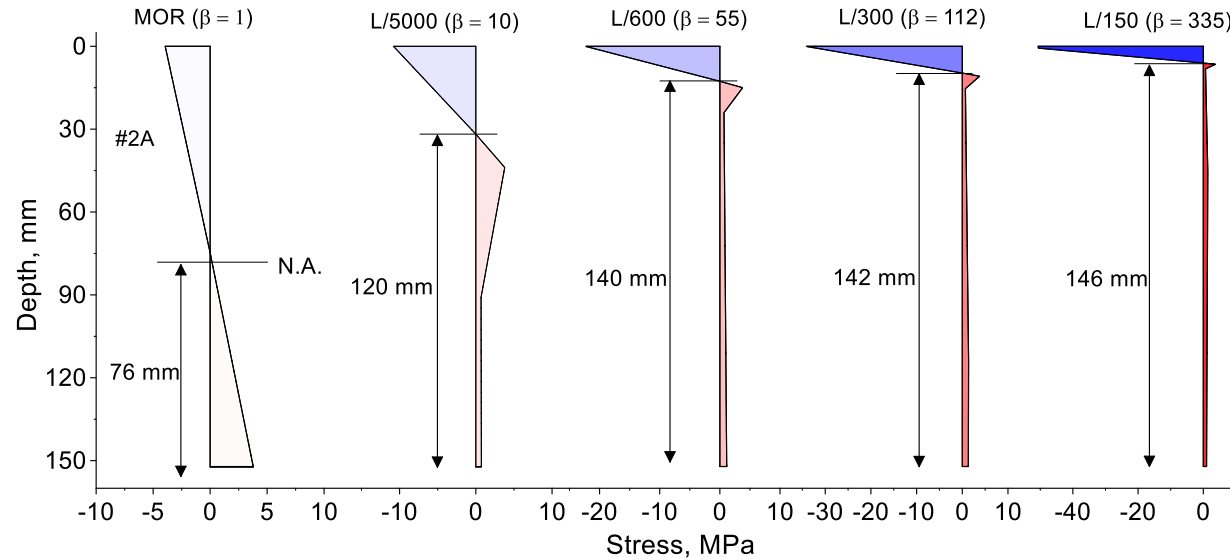
Definition of Average Tensile Stress for moment-curvature capacity in Simplified Design



Total Force in T/C $F_C = F_T = \sum_0^{(1-k)h} F_t(\varepsilon) = b \int_0^{(1-k)h} \sigma(\varepsilon) dy = bh \int_0^{\beta_i} \sigma_t(\beta_i) \left(\frac{(1-k)}{\beta_i} \right) d\beta_i$

Avg. Force in T/C $\bar{\sigma}_t = \frac{F_T}{A_t} = \frac{F_T}{(1-k)hb}$

Evolution of Stress Distribution across the thickness



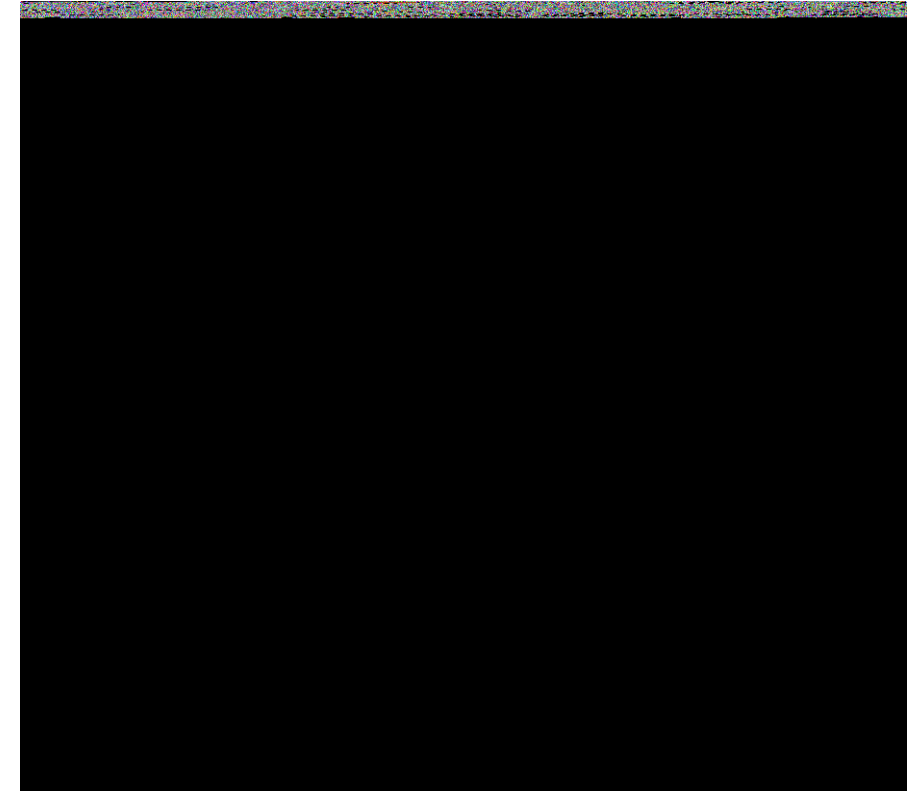
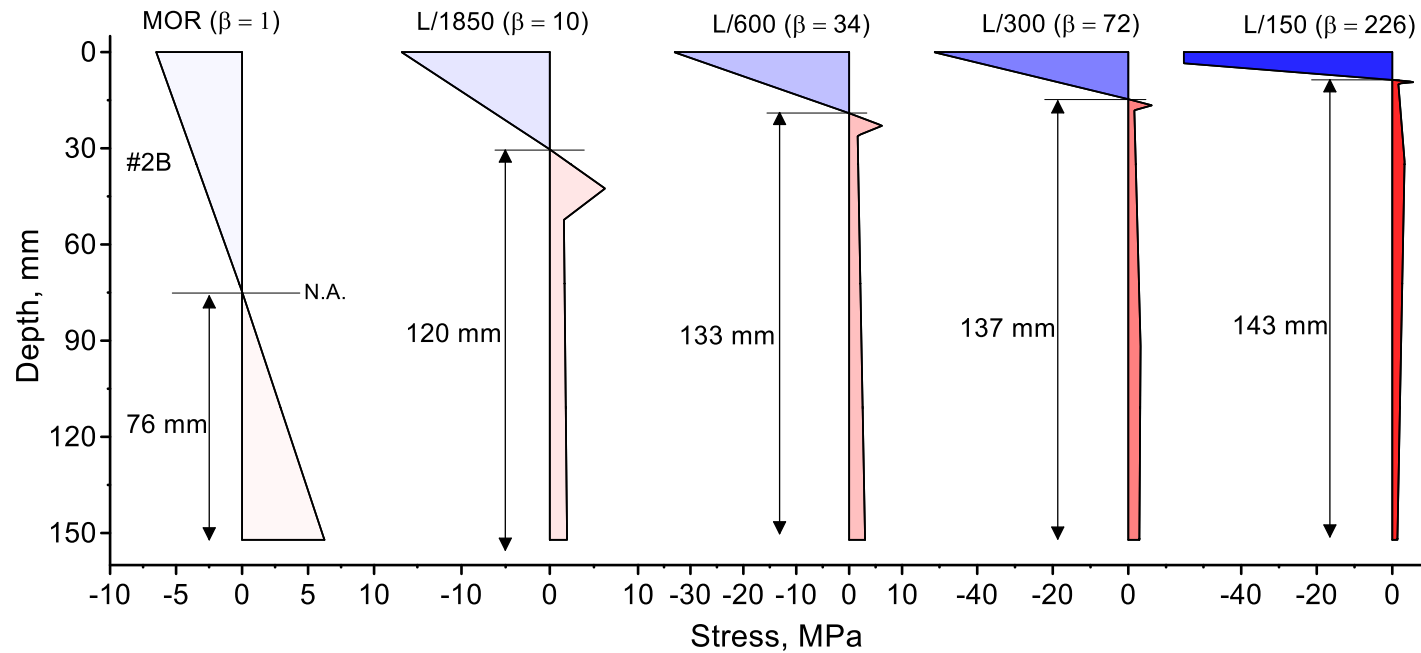
Inverse Analysis for precise characterization

- Bi-linear Model
- Trilinear Model
- Quadrilinear model
- Reinforced with Steel or FRP

Standalone Software Download Link:

<https://doi.org/10.13140/RG.2.2.12492.53127/4>

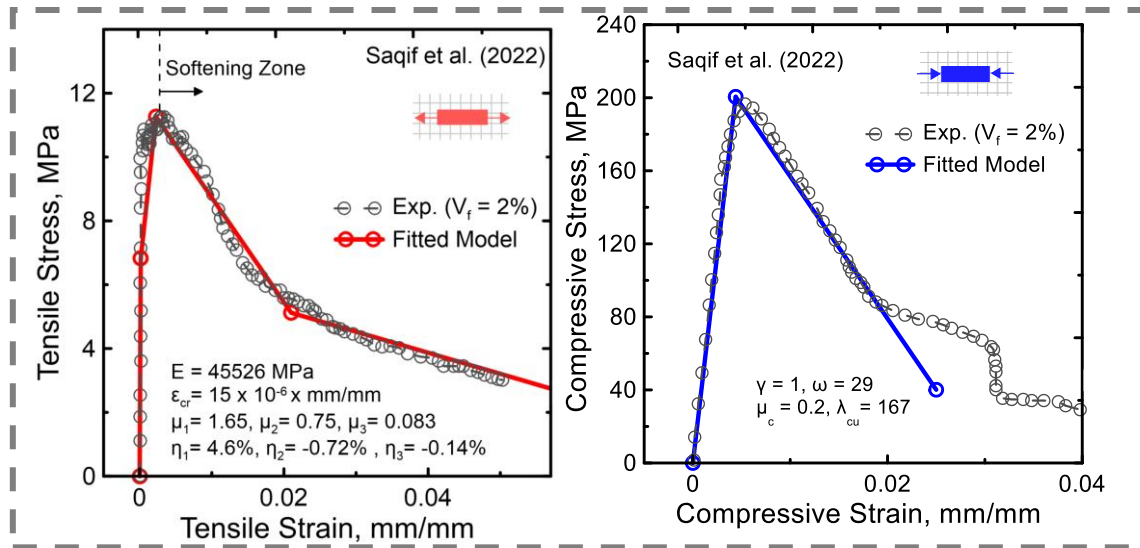
The stress limit state parameter we use for the design problem is fundamentally different



Slab on ground design using a Single value (TR 34, ACI 360, ACI 544)
vs Full response (material-based parameters analytical solution)

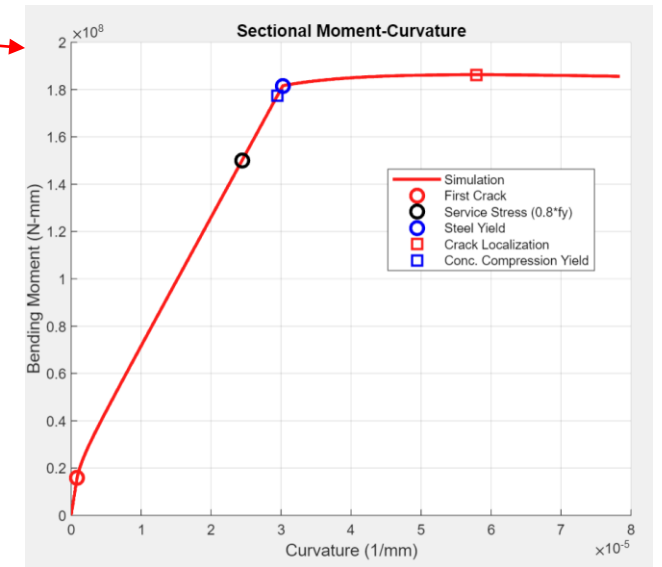
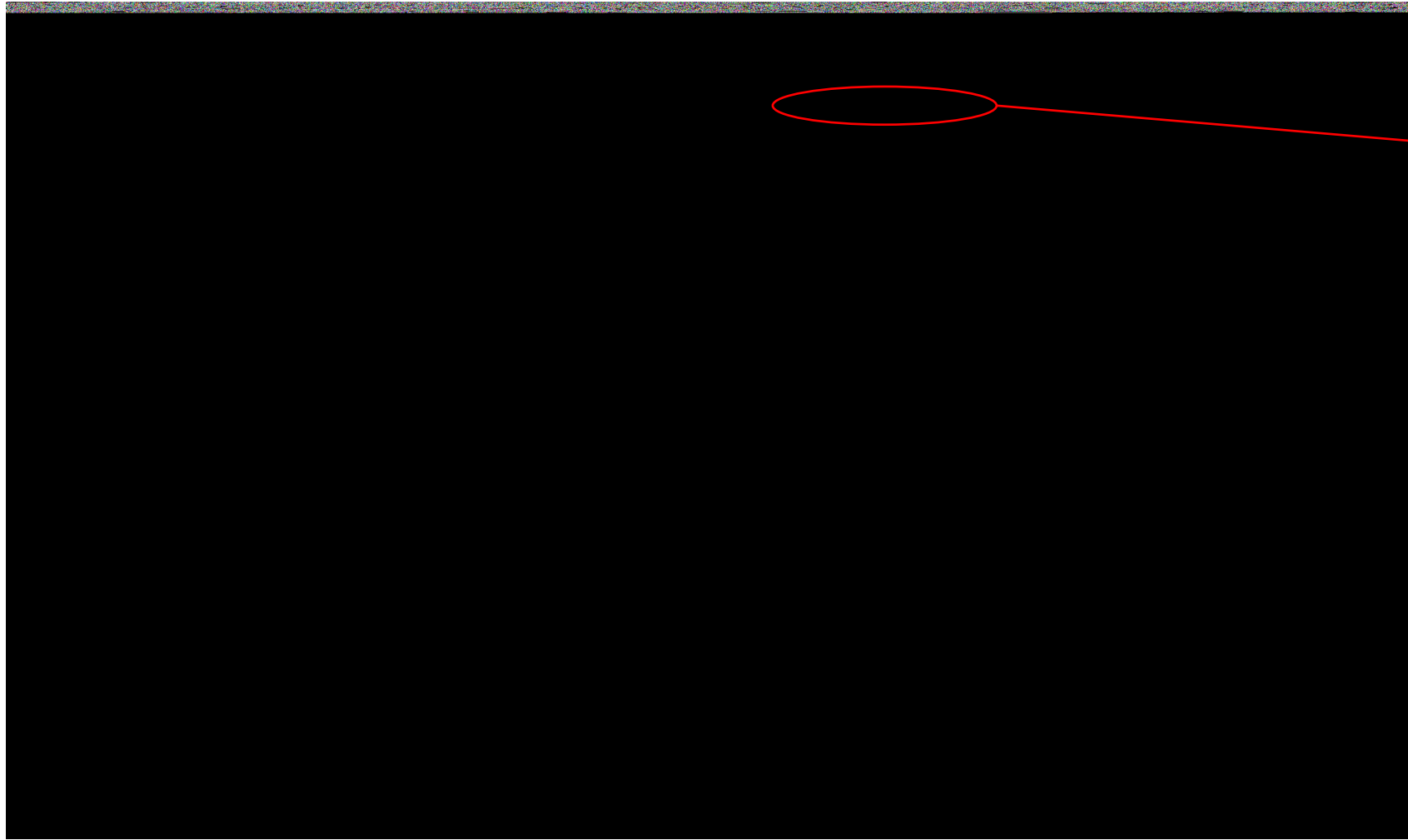
Validation of closed-form approach for Hybrid Members:

Material
Characterization



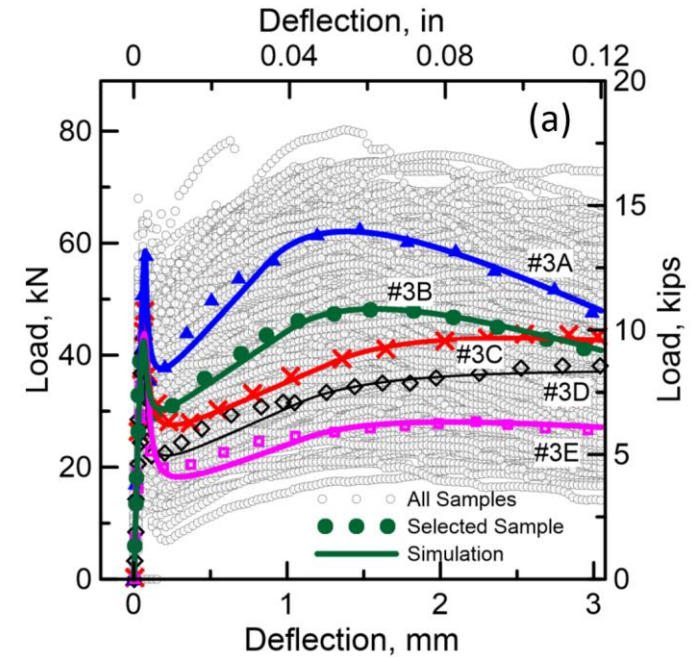
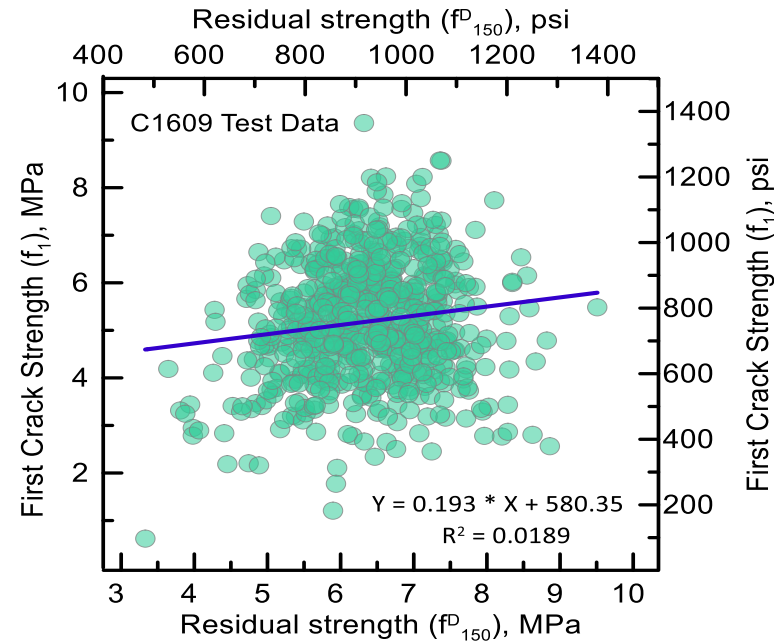
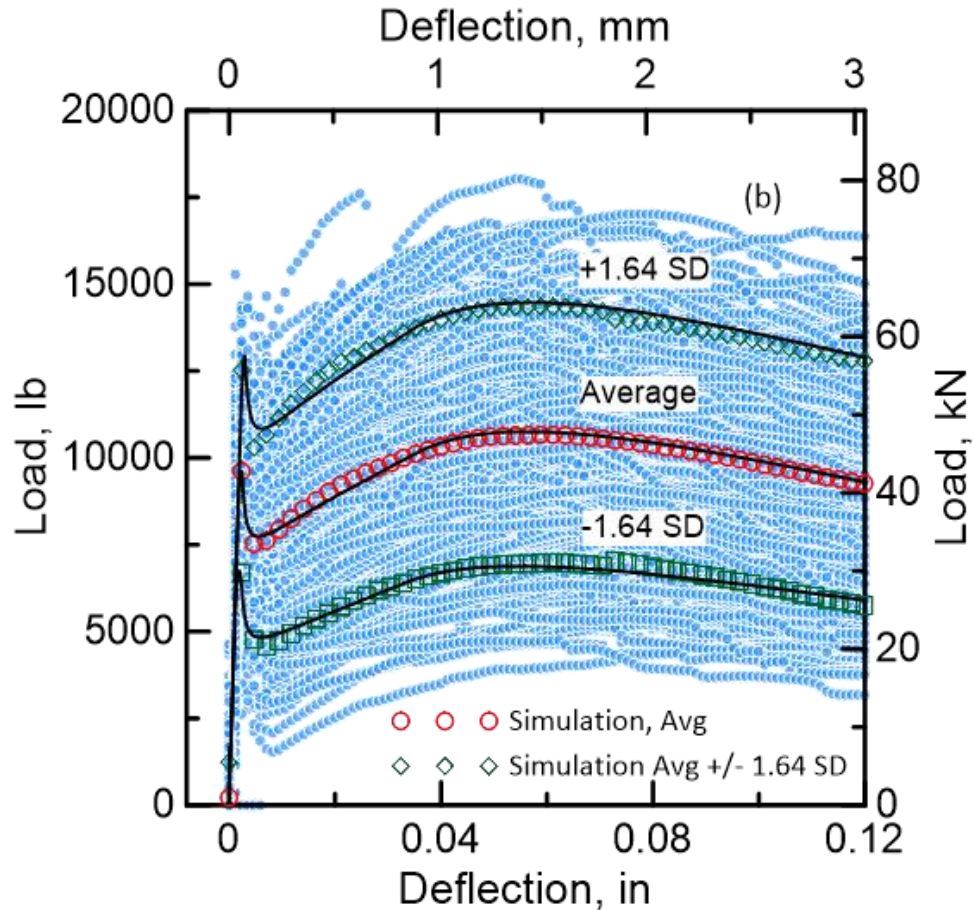
Stress Distribution
at Limit States

Design software for Plain FRC, UHPC using parametric Modeling



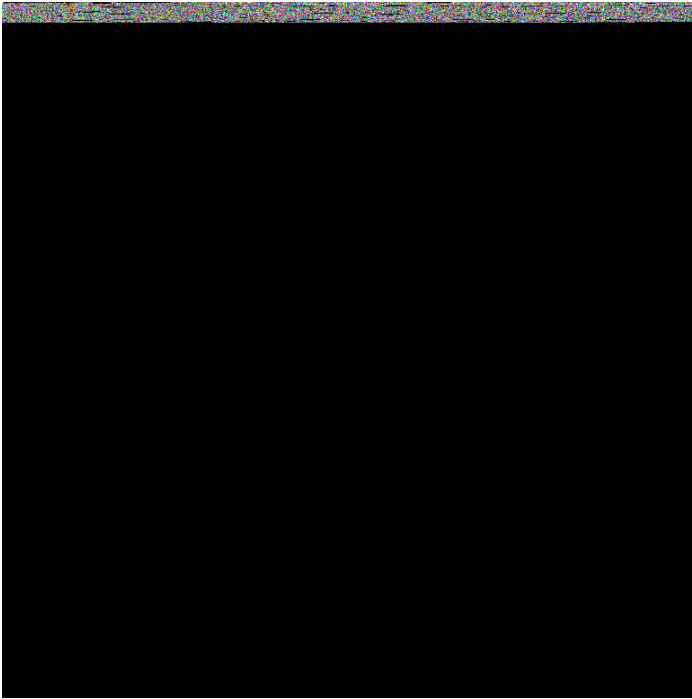
Flexural Testing used for precast Tunnel Linings QC/QA Procedures, Agentic AI applications

- Evaluation of more than 9 months of QC/QA beam samples, 1200+ flexural C1609 beams
- Address the variations using Statistical Process Control tools

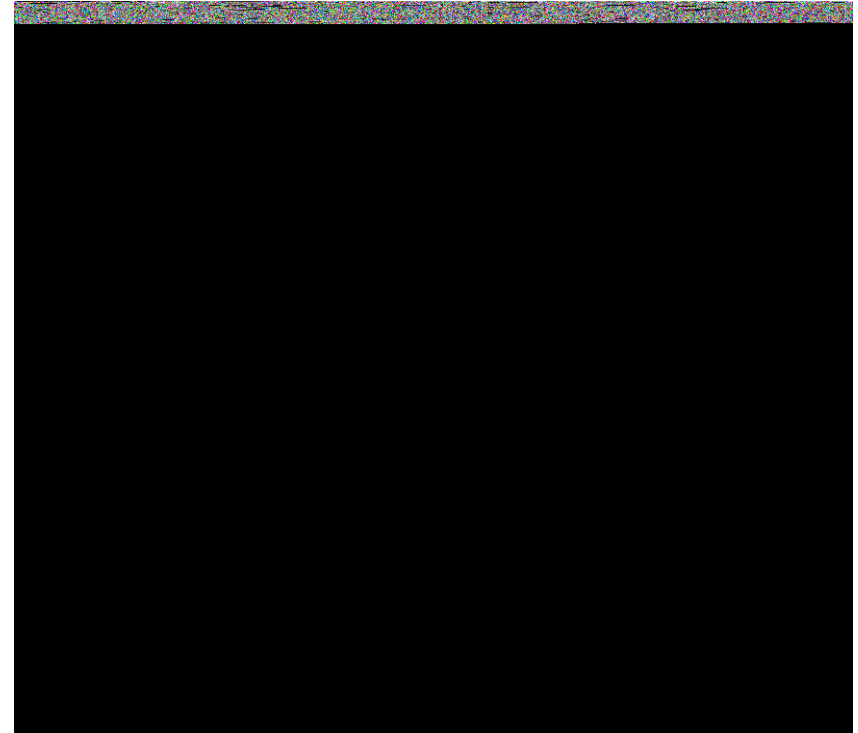


Is there any meaningful correlation?

Statistical measures of distribution and correlation

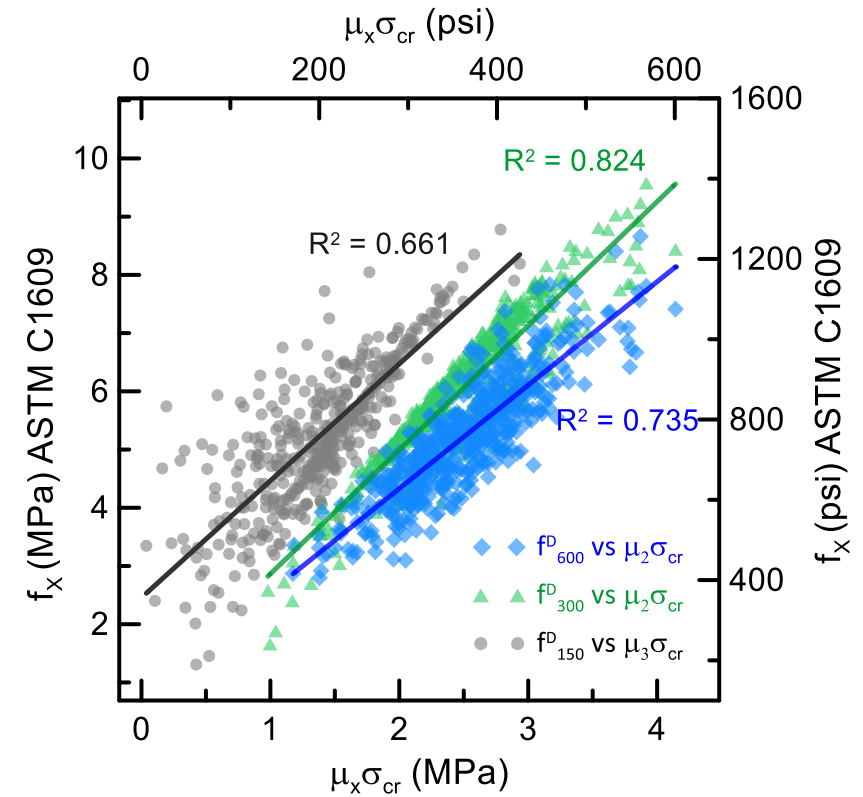
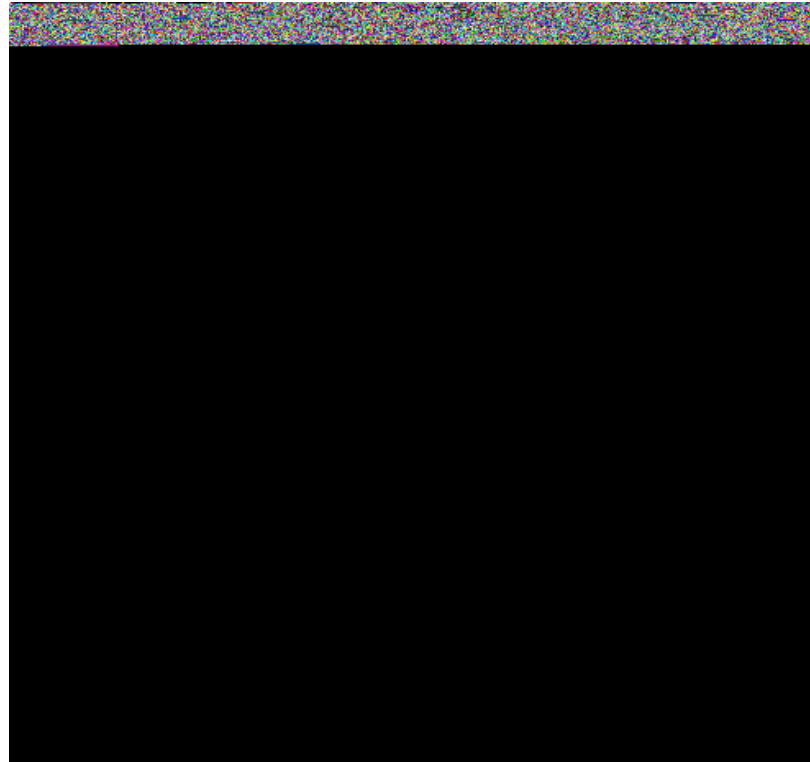
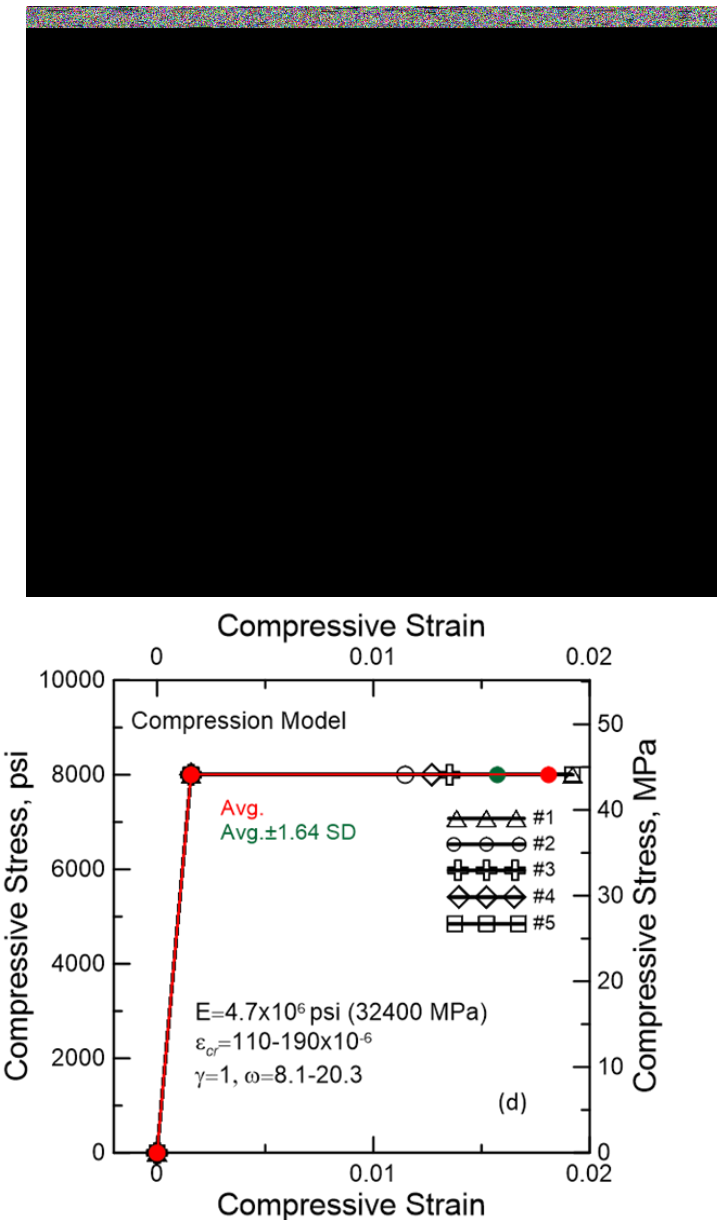


Fit Exp. Data



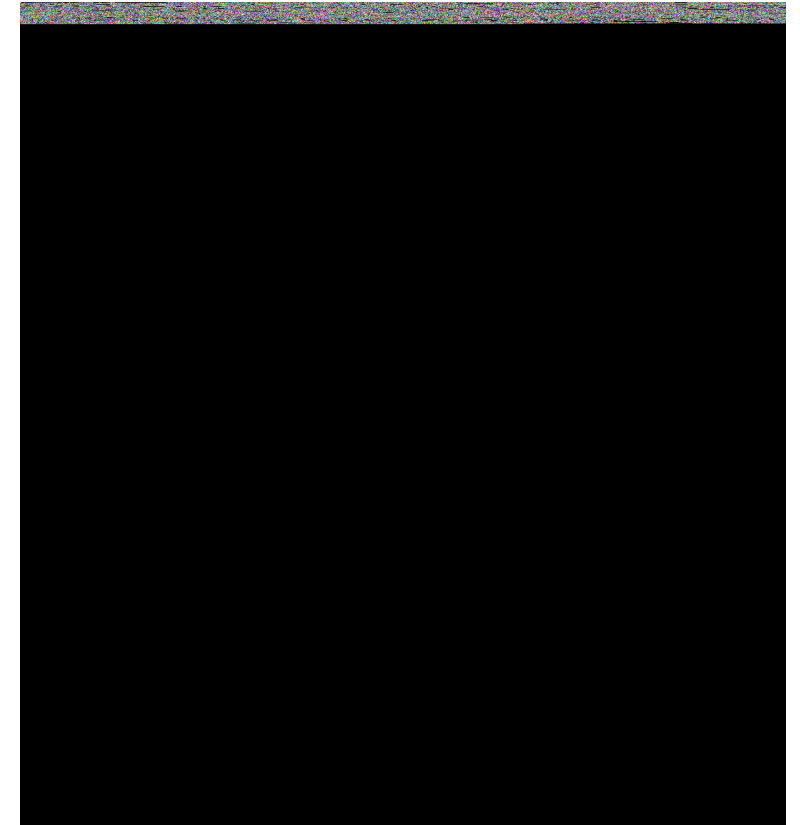
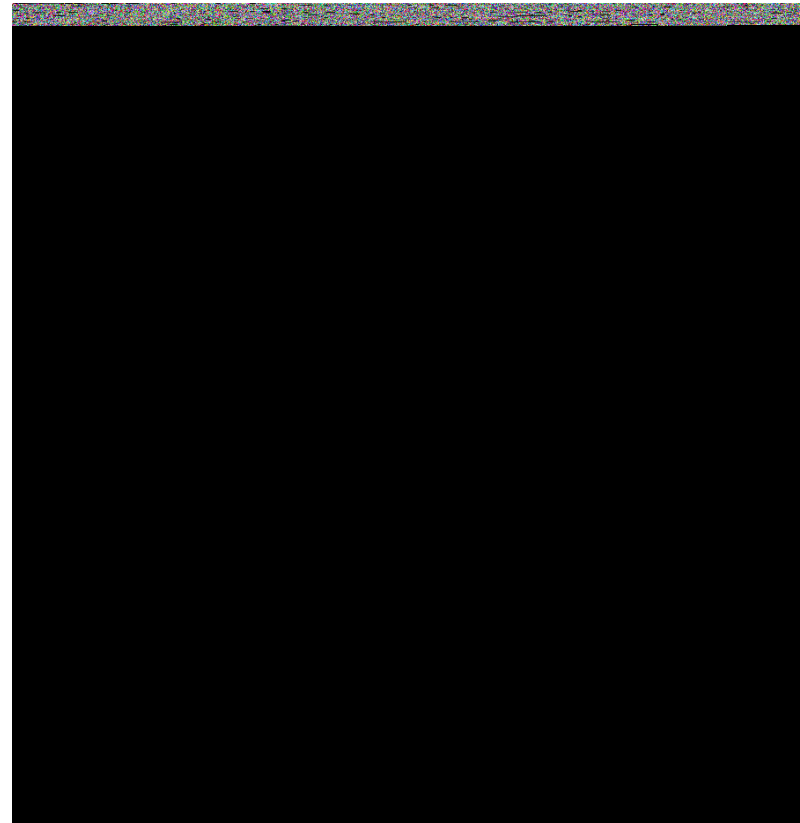
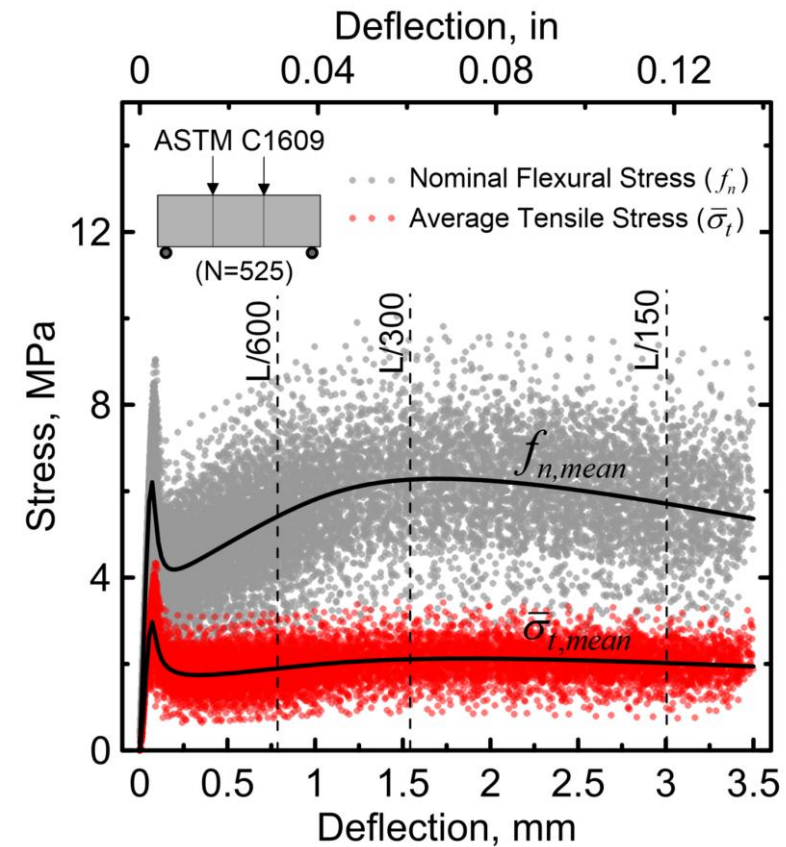
Backcalculated Tension
Model

Tensile Properties, 800+ samples

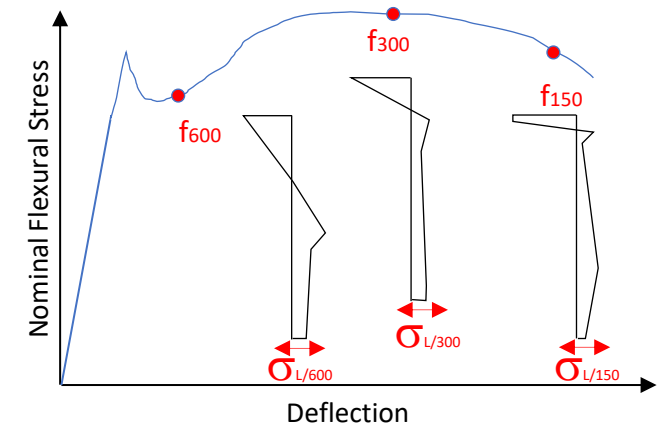
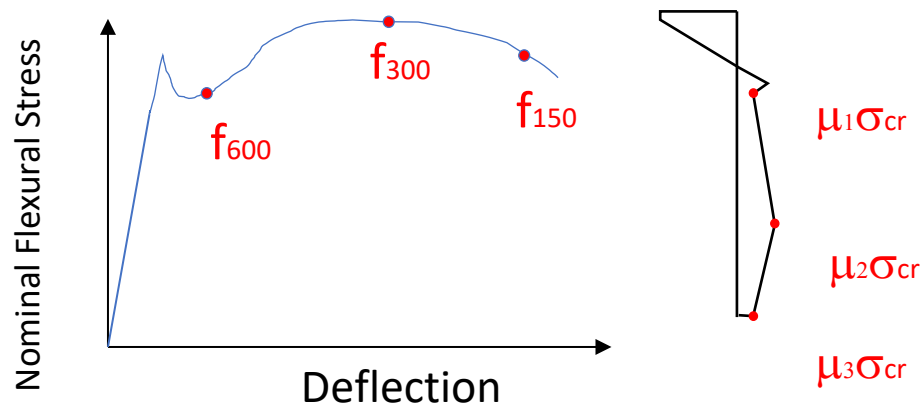
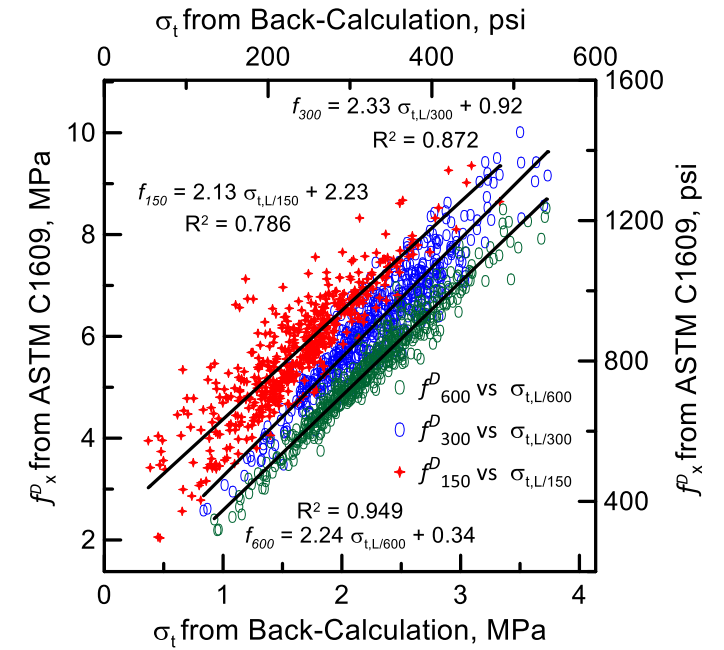
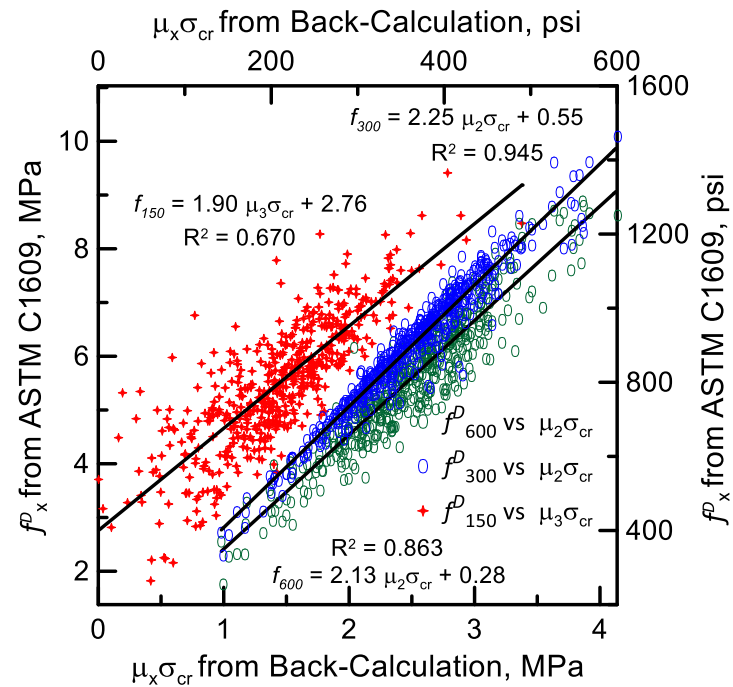


Average tensile stress calculation for simplified design

- The Nominal stress ratio and the neutral axis are relatively consistent compared to the nominal flexural stress which has a high scatter.
- The empirical factors for flexure are not constant and compound the definition of characteristic empirical factors.
- The post-peak response of different fiber types and amounts are fundamentally different.



Correlation of experimental and simulated stresses at various Deflections

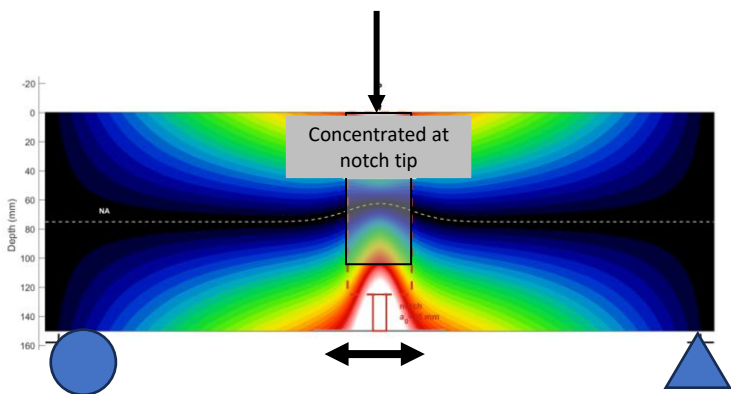
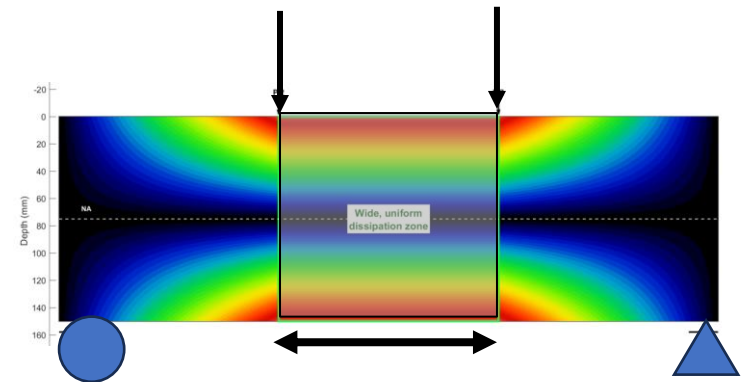


ASTMC1609 Parameters vs Tension Model Parameters

ASTMC1609 Parameters vs stress at extreme fiber

Strain-Rate Field: Notched vs. Unnotched

Gradient show relative strain-rate intensity.



Under plane-sections-remain-plane: $\dot{\epsilon}(x,z) = -z \cdot \dot{\kappa}(x)$

	ASTM C1609 (unnotched 4PB)	Notched 3PB
Active section depth	Full $h = 150$ mm	$h_{sp} = h - a_0 = 125$ mm
$d\dot{\kappa}/dx$ in active zone	= 0 (constant M zone)	≠ 0 (triangular M, EI jump at notch)
EI stiffness change	No notch induced geometric EI jump	Geometric EI reduced at notch because “h” changed
Gradient length L_V	$L/3 = 150$ mm	≈ 30 mm (at FPZ)

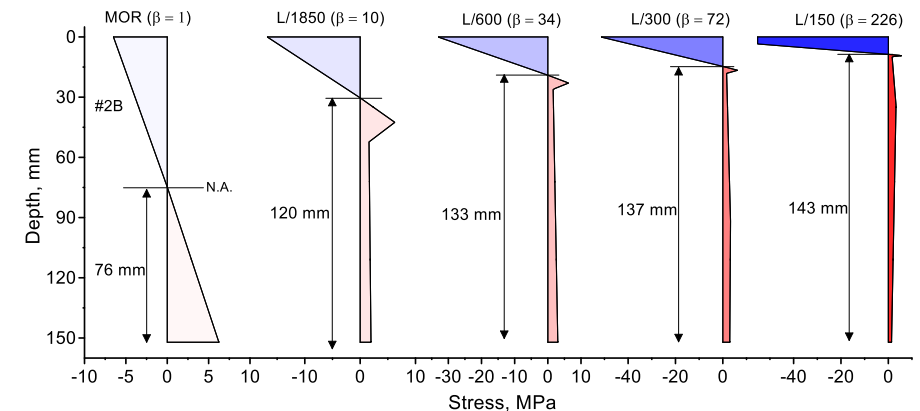
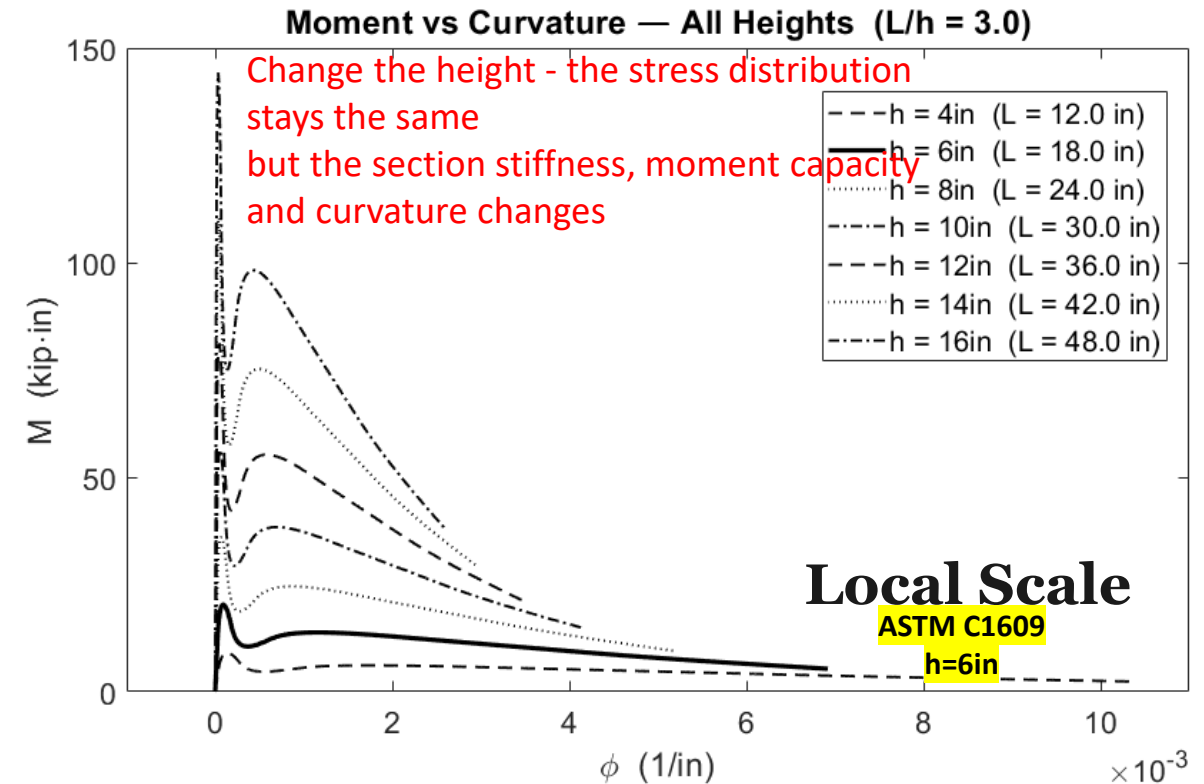
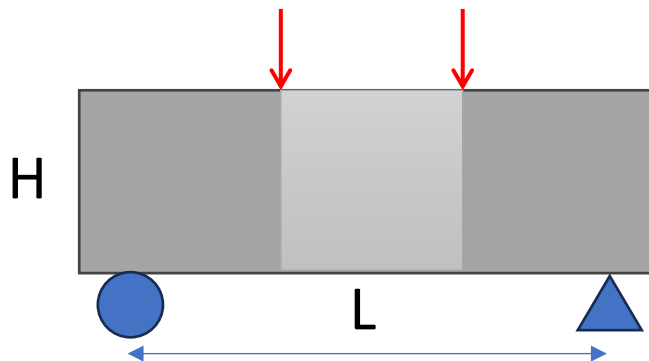
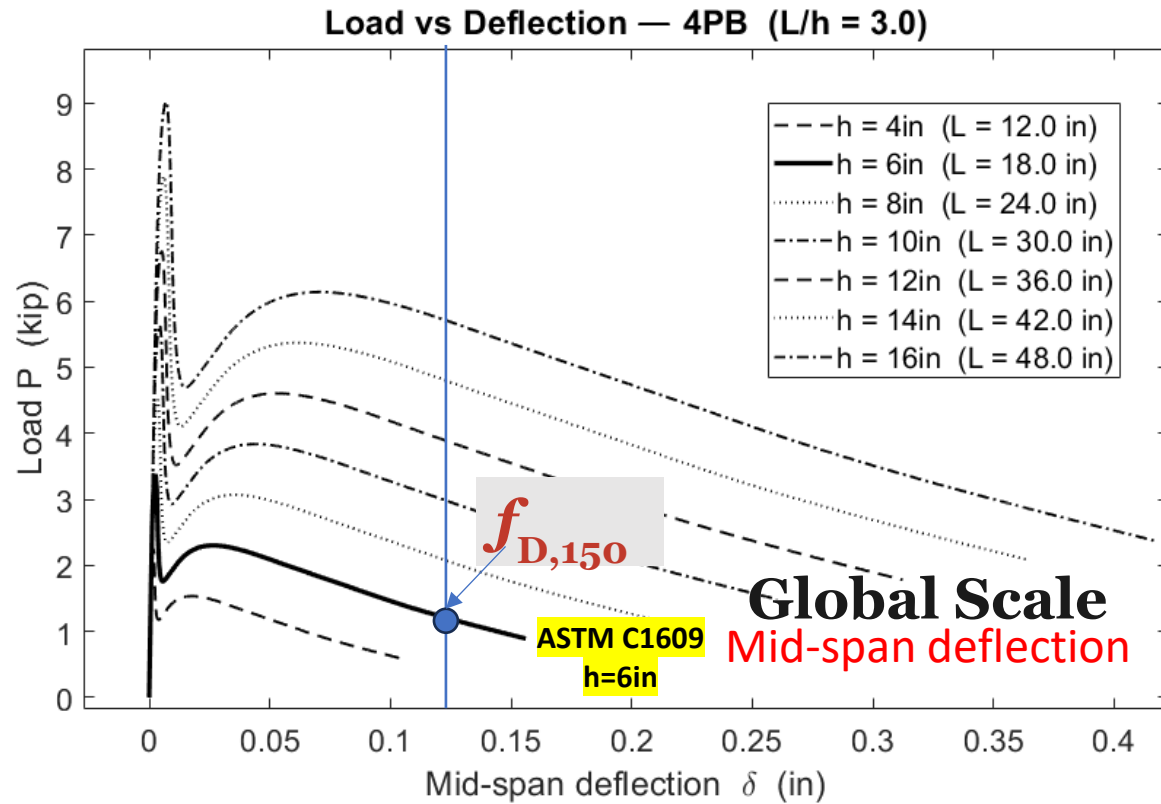
ASTM C1609 is more suitable for continuum-based scaling to full-scale beam design when the structural field remains smooth and the characteristic gradient length is much larger than the characteristic flaw spacing.

Notch 3PB should not be assumed to scale directly in the same way, because the notch localizes the response to a single ligament and increases sensitivity to the local flaw distribution

How we interpret the ASTM C1609 results

These 2 Slides explain why you need back calculation process rather than directly use the ASTM 1609 residual flexural strength

Global scale from flexural tests to Material parameter (Local Scale) of different section height and span of 4PB



What Residual flexural strength such as $f_{D,150}$ in Standard beam tells you

- $f_{D,150}$ is not a unique material constant — it changes if the specimen size, span, or boundary condition is changed. Each configuration produces a different strain gradient at the same deflection.
- Deflection is a global (structural) response, while stress distribution is a local (section-level) quantity — mapping one to the other requires knowing the full moment distribution along the span, not just the peak moment or characteristic residual moment such as $f_{D,150}$
- This is exactly why we need **a back-calculation approach** to obtain the material properties. A single point on the full load–deflection curve (e.g., $f_{D,150}$) is not enough. It captures one snapshot of one test configuration.
 - Back-calculation separates the intrinsic material behavior from the specimen geometry and loading configuration, which is essential for a complete structural design framework.

“ $f_{D,150}$ tells you what the specimen did. The moment-curvature tells you what the structure will do.”

The difference in load-deflection under closed-loop control when you control the actuator and when you control with the LVDT.

For back-calculation, you need to use the data from LVDT control before proceeding back calculation

Why Does the Load Drop at First Crack adversely affect the results ?

The ASTM C1609 Snap-Through Instability — A Machine-Specimen Interaction

What Happens

1. Before cracking

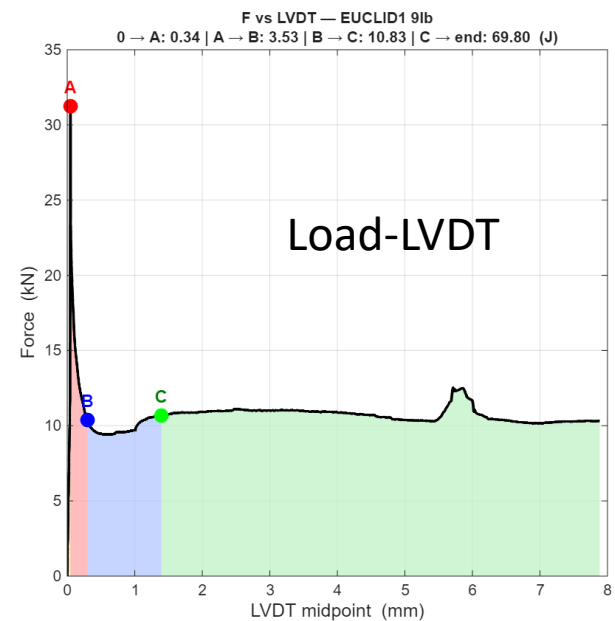
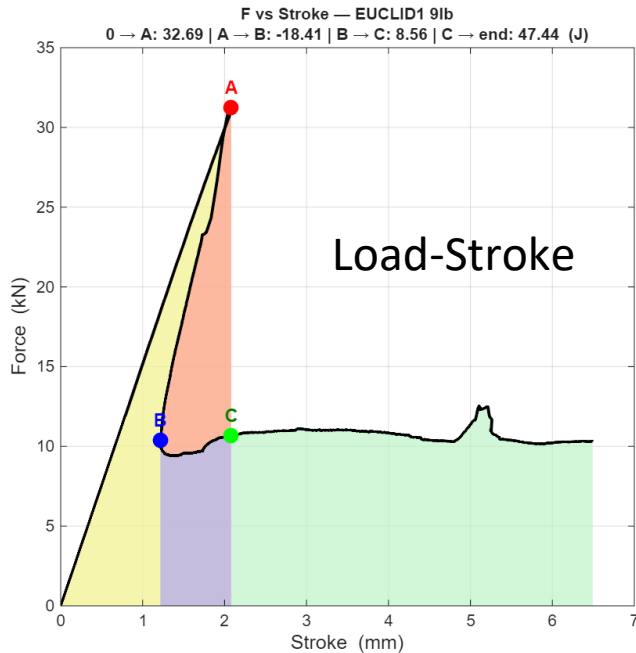
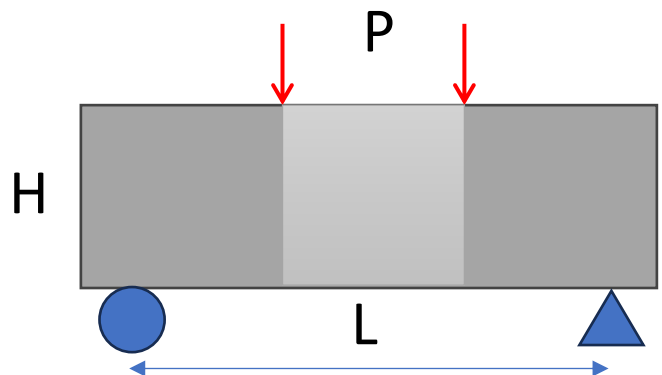
The machine frame compresses like a spring under load. It stores **97.8%** of the total elastic energy.

2. At first crack

The stiffness of beam drops suddenly.
The frame stored energy releases into the specimen driving the crack forward.

3. After the snap

Fibers bridge the crack. The test is stable again.
 $f_{D,L/600}$, $f_{D,L/300}$, $f_{D,L/300}$ are all measured in this zone.



At first crack (peak load)

Quantity	Value
Peak load	31,235 N
Stroke at peak	2.081 mm
LVDT at peak	0.046 mm
Machine deformation	2.035 mm
Energy in frame	31.78 J (97.8%)
Energy in specimen	0.72 J (2.2%)